CHAPTER 7

Alkenes: Structure and Reactivity



FIGURE 7.1 **Tomatoes are good for you**. Their red color is due to lycopene, which has 13 double bonds. (credit: modification of work "Tomatoes" by Jeremy Keith/Flickr, CC BY 2.0)

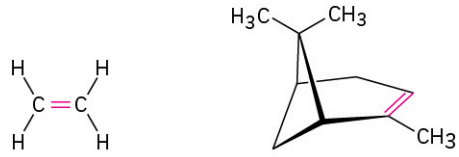
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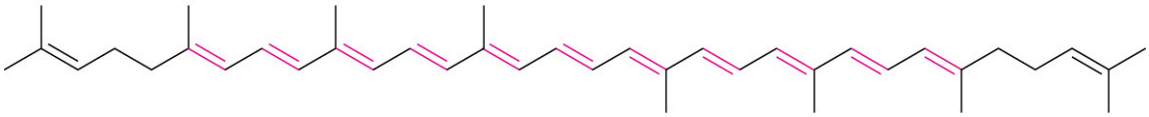
WHY THIS CHAPTER? Carbon–carbon double bonds are present in most organic and biological molecules, so a good understanding of their behavior is needed. In this chapter, we'll look at some consequences of alkene stereoisomerism and then focus on the broadest and most general class of alkene reactions, the electrophilic addition reaction. Carbon-carbon triple bonds, by contrast, occur much less commonly, so we'll not spend much time on their chemistry.

An **alkene**, sometimes called an *olefin* from the German term for oil forming, is a hydrocarbon that contains a carbon–carbon double bond, while an **alkyne** is a hydrocarbon that contains a carbon-carbon triple bond. Alkenes occur abundantly in nature. Ethylene, for instance, is a plant hormone that induces ripening in fruit, and α-pinene is the major component of turpentine. Lycopene, found in fruits such as watermelon and papaya

as well as tomatoes, is an antioxidant with numerous health benefits such as sun protection and cardiovascular protection.



Ethylene **α-Pinene**



Lycopene, a conjugated polyene

7.1 Industrial Preparation and Use of Alkenes

Ethylene and propylene, the simplest alkenes, are the two most important organic chemicals produced industrially. Approximately 220 million tons of ethylene and 138 million tons of propylene are produced worldwide each year for use in the synthesis of polyethylene, polypropylene, ethylene glycol, acetic acid, acetaldehyde, and a host of other substances (FIGURE 7.2).

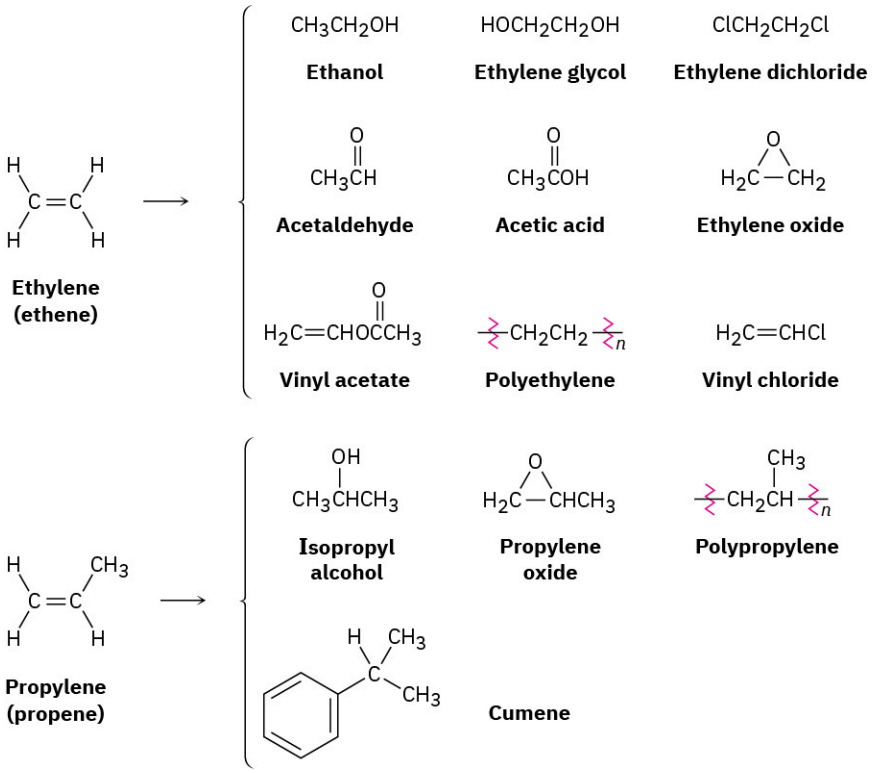


FIGURE 7.2 Compounds derived industrially from ethylene and propylene.

Ethylene, propylene, and butene are synthesized from (C_2–C_8) alkanes by a process called *steam cracking* at temperatures up to 900 °C.

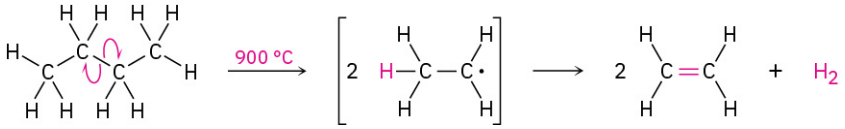
$$CH_3(CH_2)_nCH_3 \quad [n = 0\text{–}6]$$

$$\downarrow \text{ 850–900 °C, steam}$$

$$H_2 + H_2C{=}CH_2 + CH_3CH{=}CH_2 + CH_3CH_2CH{=}CH_2$$

The cracking process is complex, although it undoubtedly involves radical reactions. The high-temperature

reaction conditions cause spontaneous breaking of C–C and C–H bonds, with the resultant formation of smaller fragments. We might imagine, for instance, that a molecule of butane splits into two ethyl radicals, each of which then loses a hydrogen atom to generate two molecules of ethylene.



Steam cracking is an example of a reaction whose energetics are dominated by entropy ($\Delta S°$) rather than by enthalpy ($\Delta H°$) in the free-energy equation $\Delta G° = \Delta H° - T\Delta S°$ discussed in **Section 6.7**. Although the bond dissociation energy D for a carbon–carbon single bond is relatively high (about 370 kJ/mol) and cracking is endothermic, the large positive entropy change resulting from the fragmentation of one large molecule into several smaller pieces, together with the high temperature, makes the $T\Delta S°$ term larger than the $\Delta H°$ term, thereby favoring the cracking reaction.

7.2 Calculating the Degree of Unsaturation

Because of its double bond, an alkene has fewer hydrogens than an alkane with the same number of carbons—C_nH_{2n} for an alkene versus C_nH_{2n+2} for an alkane—and is therefore referred to as **unsaturated**. Ethylene, for example, has the formula C_2H_4, whereas ethane has the formula C_2H_6.

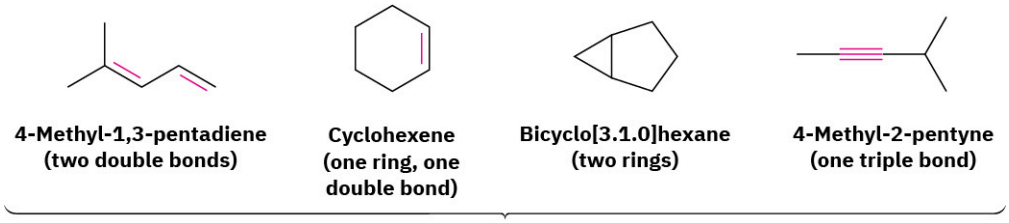


Ethylene: C_2H_4
(Fewer hydrogens—*Unsaturated*)

Ethane: C_2H_6
(More hydrogens—*Saturated*)

In general, each ring or double bond in a molecule corresponds to a loss of two hydrogens from the alkane formula C_nH_{2n+2}. Knowing this relationship, it's possible to work backward from a molecular formula to calculate a molecule's **degree of unsaturation**—the number of rings and/or multiple bonds present in the molecule.

Let's assume that we want to find the structure of an unknown hydrocarbon. A molecular weight determination yields a value of 82 amu, which corresponds to a molecular formula of C_6H_{10}. Since the saturated C_6 alkane (hexane) has the formula C_6H_{14}, the unknown compound has two fewer pairs of hydrogens ($H_{14} - H_{10} = H_4 = 2\ H_2$) so its degree of unsaturation is 2. The unknown therefore contains either two double bonds, one ring and one double bond, two rings, or one triple bond. There's still a long way to go to establish its structure, but the simple calculation has told us a lot about the molecule.



4-Methyl-1,3-pentadiene (two double bonds)

Cyclohexene (one ring, one double bond)

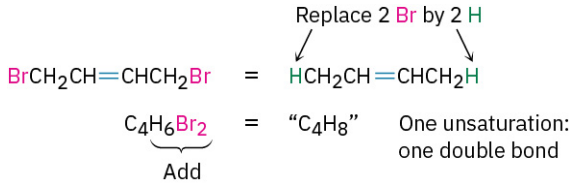
Bicyclo[3.1.0]hexane (two rings)

4-Methyl-2-pentyne (one triple bond)

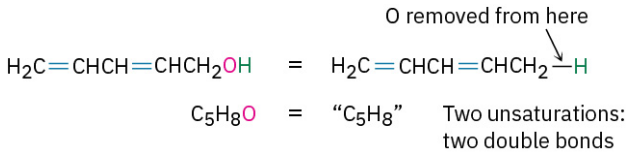
C_6H_{10}

Similar calculations can be carried out for compounds containing elements other than just carbon and hydrogen.

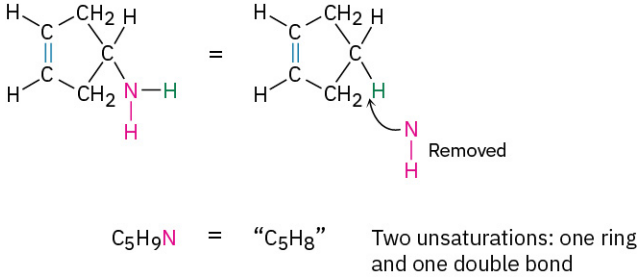
- **Organohalogen compounds (C, H, X, where X = F, Cl, Br, or I)** A halogen substituent acts as a replacement for hydrogen in an organic molecule, so we can add the number of halogens and hydrogens to arrive at an equivalent hydrocarbon formula from which the degree of unsaturation can be found. For example, the formula $C_4H_6Br_2$ is equivalent to the hydrocarbon formula C_4H_8 and thus corresponds to one degree of unsaturation.



- **Organooxygen compounds (C, H, O)** Oxygen forms two bonds, so it doesn't affect the formula of an equivalent hydrocarbon and can be ignored when calculating the degree of unsaturation. You can convince yourself of this by seeing what happens when an oxygen atom is inserted into an alkane bond: C–C becomes C–O–C or C–H becomes C–O–H, and there is no change in the number of hydrogen atoms. For example, the formula C_5H_8O is equivalent to the hydrocarbon formula C_5H_8 and thus corresponds to two degrees of unsaturation.



- **Organonitrogen compounds (C, H, N)** Nitrogen forms three bonds, so an organonitrogen compound has one more hydrogen than a related hydrocarbon. We therefore subtract the number of nitrogens from the number of hydrogens to arrive at the equivalent hydrocarbon formula. Again, you can convince yourself of this by seeing what happens when a nitrogen atom is inserted into an alkane bond: C–C becomes C–NH–C or C–H becomes C–NH_2, meaning that one additional hydrogen atom has been added. We must therefore subtract this extra hydrogen atom to arrive at the equivalent hydrocarbon formula. For example, the formula C_5H_9N is equivalent to C_5H_8 and thus has two degrees of unsaturation.



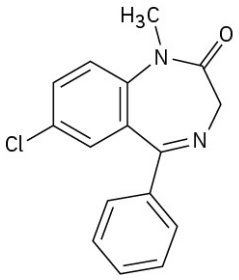
To summarize:

- **Add** the number of halogens to the number of hydrogens.
- **Ignore** the number of oxygens.
- **Subtract** the number of nitrogens from the number of hydrogens.

PROBLEM 7-1 Calculate the degree of unsaturation in each of the following formulas, and then draw as many structures as you can for each:
(a) C_4H_8 **(b)** C_4H_6 **(c)** C_3H_4

PROBLEM 7-2 Calculate the degree of unsaturation in each of the following formulas:
(a) C_6H_5N **(b)** $C_6H_5NO_2$ **(c)** $C_8H_9Cl_3$ **(d)** $C_9H_{16}Br_2$ **(e)** $C_{10}H_{12}N_2O_3$ **(f)** $C_{20}H_{32}ClN$

PROBLEM 7-3 Diazepam, marketed as an antianxiety medication under the name Valium, has three rings, eight double bonds, and the formula $C_{16}H_?ClN_2O$. How many hydrogens does diazepam have? (Calculate the answer; don't count hydrogens in the structure.)



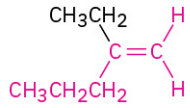
Diazepam

7.3 Naming Alkenes

Alkenes are named using a series of rules similar to those for alkanes (**Section 3.4**), with the suffix *-ene* used instead of *-ane* to identify the functional group. There are three steps to this process.

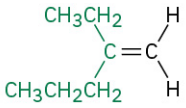
STEP 1

Name the parent hydrocarbon. Find the longest carbon chain containing the double bond, and name the compound accordingly, using the suffix *-ene:*



Named as a *pentene*

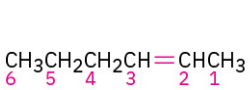
NOT

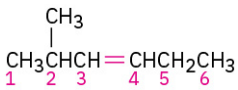


as a hexene, since the double bond is not contained in the six-carbon chain

STEP 2

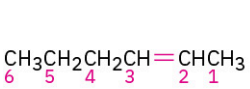
Number the carbon atoms in the chain. Begin at the end nearer the double bond or, if the double bond is equidistant from the two ends, begin at the end nearer the first branch point. This rule ensures that the double-bond carbons receive the lowest possible numbers.



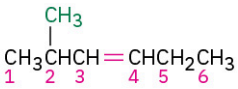


STEP 3

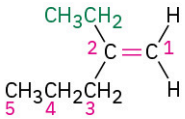
Write the full name. Number the substituents according to their positions in the chain, and list them alphabetically. Indicate the position of the double bond by giving the number of the first alkene carbon and placing that number directly before the parent name. If more than one double bond is present, indicate the position of each and use one of the suffixes *-diene, -triene,* and so on.



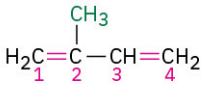
2-Hexene



2-Methyl-3-hexene

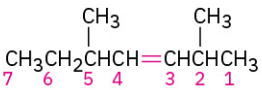


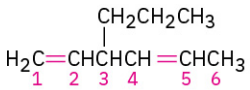
2-Ethyl-1-pentene



2-Methyl-1,3-butadiene

We should also note that IUPAC changed their naming recommendations in 1993 to place the locant indicating the position of the double bond immediately before the *-ene* suffix rather than before the parent name: but-2-ene rather than 2-butene, for instance. This change has not been widely accepted by the chemical community in the United States, however, so we'll stay with the older but more commonly used names. Be aware, though, that you may occasionally encounter the newer system.

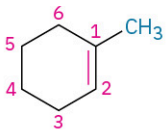


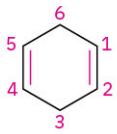


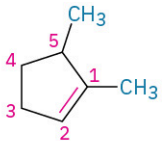
Newer naming system: **2,5-Dimethylhept-3-ene** **3-Propylhexa-1,4-diene**

(Older naming system: **2,5-Dimethyl-3-heptene** **3-Propyl-1,4-hexadiene)**

Cycloalkenes are named similarly, but because there is no chain end to begin from, we number the cycloalkene so that the double bond is between C1 and C2 and the first substituent has as low a number as possible. It's not necessary to indicate the position of the double bond in the name because it's always between C1 and C2. As with open-chain alkenes, the newer but not yet widely accepted naming rules place the locant immediately before the suffix in a cyclic alkene.



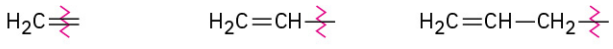




1-Methylcyclohexene **1,4-Cyclohexadiene** **1,5-Dimethylcyclopentene**

(New: Cyclohexa-1,4-diene)

For historical reasons, there are a few alkenes whose names are firmly entrenched in common usage but don't conform to the rules. For example, the alkene derived from ethane should be called *ethene,* but the name *ethylene* has been used for so long that it is accepted by IUPAC. TABLE 7.1 lists several other common names that are often used and are recognized by IUPAC. Note also that a $=CH_2$ substituent is called a **methylene group**, a $H_2C{=}CH-$ substituent is called a **vinyl group**, and a $H_2C{=}CHCH_2-$ substituent is called an **allyl group**.

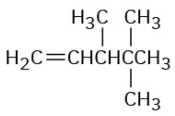


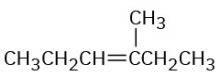
A methylene group **A vinyl group** **An allyl group**

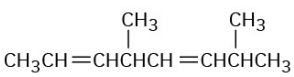
TABLE 7.1 Common Names of Some Alkenes

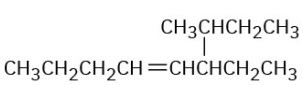
Compound	Systematic name	Common name
$H_2C{=}CH_2$	Ethene	Ethylene
$CH_3CH{=}CH_2$	Propene	Propylene
$CH_3C(CH_3){=}CH_2$	2-Methylpropene	Isobutylene
$H_2C{=}C(CH_3)-CH{=}CH_2$	2-Methyl-1,3-butadiene	Isoprene

PROBLEM 7-4 Give IUPAC names for the following compounds:

(a) 

(b) 

(c) 

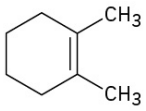
(d) 

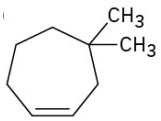
PROBLEM 7-5 Draw structures corresponding to the following IUPAC names:

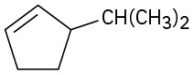
(a) 2-Methyl-1,5-hexadiene **(b)** 3-Ethyl-2,2-dimethyl-3-heptene

(c) 2,3,3-Trimethyl-1,4,6-octatriene **(d)** 3,4-Diisopropyl-2,5-dimethyl-3-hexene

PROBLEM 7-6 Name the following cycloalkenes:

(a) 

(b) 

(c) 

PROBLEM 7-7 Change the following old names to new, post-1993 names, and draw the structure of each compound:

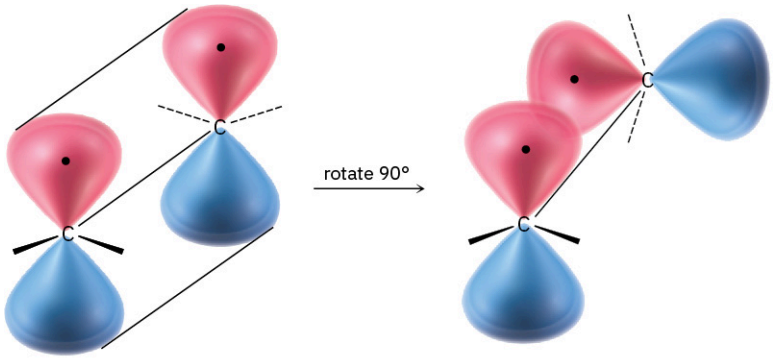
(a) 2,5,5-Trimethyl-2-hexene **(b)** 2,3-Dimethyl-1,3-cyclohexadiene

7.4 Cis–Trans Isomerism in Alkenes

We saw in the chapter on **Structure and Bonding** that the carbon–carbon double bond can be described in two ways. In valence bond language (**Section 1.8**), the carbons are sp^2-hybridized and have three equivalent hybrid orbitals that lie in a plane at angles of 120° to one another. The carbons form a σ bond by a head-on overlap of sp^2 orbitals and form a π bond by sideways overlap of unhybridized p orbitals oriented perpendicular to the sp^2 plane, as shown in **FIGURE 1.15**.

In molecular orbital language (**Section 1.11**), interaction between the p orbitals leads to one bonding and one antibonding π molecular orbital. The π bonding MO has no node between nuclei and results from a combination of p orbital lobes with the same algebraic sign. The π antibonding MO has a node between nuclei and results from a combination of lobes with different algebraic signs, as shown in **FIGURE 1.19**.

Although essentially free rotation around single bonds is possible (**Section 3.6**), the same is not true of double bonds. For rotation to occur around a double bond, the π bond must break and re-form (**FIGURE 7.3**). Thus, the barrier to double-bond rotation must be at least as great as the strength of the π bond itself, an estimated 350 kJ/mol (84 kcal/mol). Recall that the barrier to bond rotation in ethane is only 12 kJ/mol.

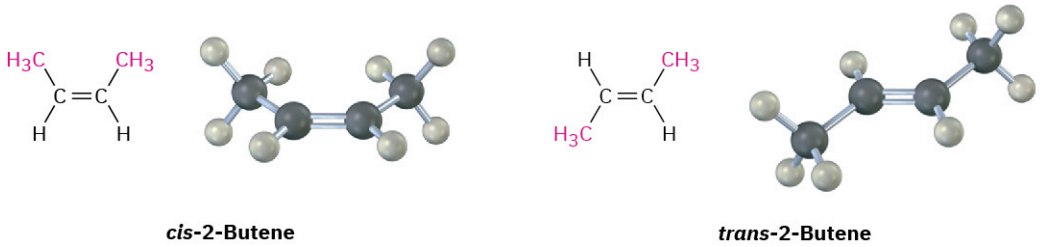


π bond (p orbitals are parallel) **Broken π bond after rotation (p orbitals are perpendicular)**

FIGURE 7.3 The π bond must break for rotation to take place around a carbon–carbon double bond.

The lack of rotation around carbon–carbon double bonds is of more than just theoretical interest; it also has chemical consequences. Imagine the situation for a disubstituted alkene such as 2-butene. (Disubstituted means that two substituents other than hydrogen are bonded to the double-bond carbons.) The two methyl groups in 2-butene can either be on the same side of the double bond or on opposite sides, a situation similar to that in disubstituted cycloalkanes (**Section 4.2**).

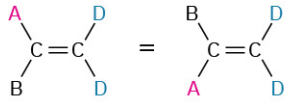
Since bond rotation can't occur, the two 2-butenes can't spontaneously interconvert; they are different, isolable compounds. As with disubstituted cycloalkanes, we call such compounds cis–trans stereoisomers. The compound with substituents on the same side of the double bond is called *cis*-2-butene, and the isomer with substituents on opposite sides is *trans*-2-butene (**FIGURE 7.4**).



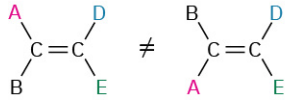
***cis*-2-Butene** ***trans*-2-Butene**

FIGURE 7.4 Cis and trans isomers of 2-butene. The cis isomer has the two methyl groups on the same side of the double bond, and the trans isomer has methyl groups on opposite sides.

Cis–trans isomerism is not limited to disubstituted alkenes. It can occur whenever both double-bond carbons are attached to two different groups. If one of the double-bond carbons is attached to two identical groups, however, cis–trans isomerism is not possible (**FIGURE 7.5**).



These two compounds are identical; they are not cis–trans isomers.



These two compounds are not identical; they are cis–trans isomers.

FIGURE 7.5 The requirement for cis–trans isomerism in alkenes. Compounds that have one of their carbons bonded to two identical groups can't exist as cis–trans isomers. Only when both carbons are bonded to two different groups is cis–trans isomerism possible.

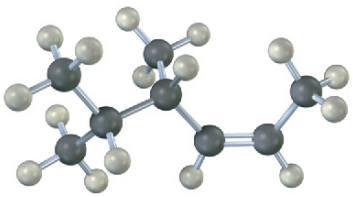
PROBLEM 7-8 The sex attractant of the common housefly is an alkene named *cis*-9-tricosene. Draw its structure. (Tricosane is the straight-chain alkane $C_{23}H_{48}$.)

PROBLEM 7-9 Which of the following compounds can exist as pairs of cis–trans isomers? Draw each pair, and indicate the geometry of each isomer.

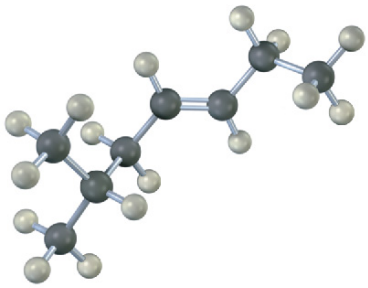
(a) $CH_3CH{=}CH_2$ **(b)** $(CH_3)_2C{=}CHCH_3$ **(c)** $CH_3CH_2CH{=}CHCH_3$ **(d)** $(CH_3)_2C{=}C(CH_3)CH_2CH_3$
(e) $ClCH{=}CHCl$ **(f)** $BrCH{=}CHCl$

PROBLEM 7-10 Name the following alkenes, including a cis or trans designation:

(a)



(b)



7.5 Alkene Stereochemistry and the *E*,*Z* Designation

The cis–trans naming system used in the previous section works only with disubstituted alkenes—compounds that have two substituents other than hydrogen on the double bond. With trisubstituted and tetrasubstituted double bonds, a more general method is needed for describing double-bond geometry. (*Trisubstituted* means three substituents other than hydrogen on the double bond; *tetrasubstituted* means four substituents other than hydrogen.)

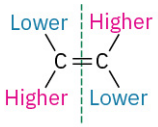
The method used for describing alkene stereochemistry is called the ***E*,*Z* system** and employs the same Cahn–Ingold–Prelog sequence rules given in **Section 5.5** for specifying the configuration of a chirality center. Let's briefly review the sequence rules and then see how they're used to specify double-bond geometry. For a more thorough review, reread **Section 5.5**.

RULE 1
Considering each of the double-bond carbons separately, look at the two substituents attached and rank them according to the atomic number of the first atom in each (8 for oxygen, 6 for carbon, 1 for hydrogen, and so forth). An atom with higher atomic number ranks higher than an atom with lower atomic number.

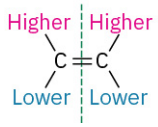
RULE 2
If a decision can't be reached by ranking the first atoms in the two substituents, look at the second, third, or fourth atoms away from the double-bond until the first difference is found.

RULE 3
Multiple-bonded atoms are equivalent to the same number of single-bonded atoms.

Once the two groups attached to each double-bonded carbon have been ranked as either higher or lower, look at the entire molecule. If the higher-ranked groups on each carbon are on the same side of the double bond, the alkene is said to have a ***Z* configuration**, for the German *zusammen,* meaning "together." If the higher-ranked groups are on opposite sides, the alkene has an ***E* configuration**, for the German *entgegen,* meaning "opposite." (For a simple way to remember which is which, note that the groups are on "ze zame zide" in the *Z* isomer.)

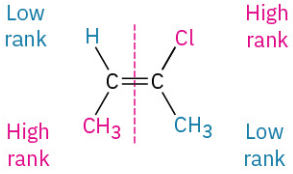


E double bond
(Higher-ranked groups
are on opposite sides.)

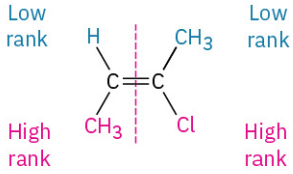


Z double bond
(Higher-ranked groups
are on the same side.)

As an example, look at the following two isomers of 2-chloro-2-butene. Because chlorine has a higher atomic number than carbon, a –Cl substituent is ranked higher than a $-CH_3$ group. Methyl is ranked higher than hydrogen, however, so isomer **(a)** is designated *E* because the higher-ranked groups are on opposite sides of the double bond. Isomer **(b)** has a *Z* configuration because its higher-ranked groups are on ze zame zide of the double bond.

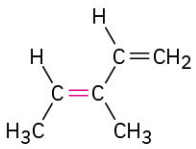


(a) (*E*)-2-Chloro-2-butene

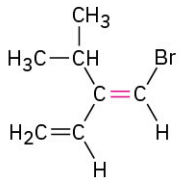


(b) (*Z*)-2-Chloro-2-butene

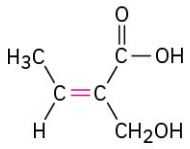
For further practice, work through each of the following examples to convince yourself that the assignments are correct:



(*E*)-3-Methyl-1,3-pentadiene



(*E*)-1-Bromo-2-isopropyl-1,3-butadiene

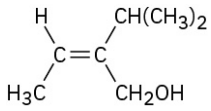


(*Z*)-2-Hydroxymethyl-2-butenoic acid

WORKED EXAMPLE 7.1

Assigning *E* and *Z* Configurations to Alkenes

Assign *E* or *Z* configuration to the double bond in the following compound:

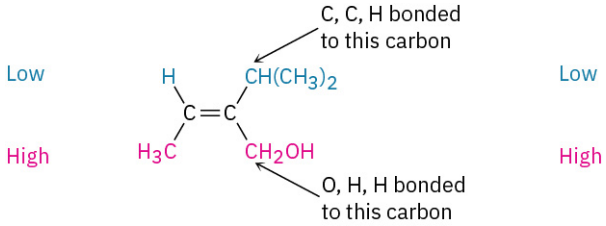


Strategy

Look at the two substituents connected to each double-bonded carbon, and determine their ranking using the Cahn–Ingold–Prelog rules. Then, check whether the two higher-ranked groups are on the same or opposite sides of the double bond.

Solution

The left-hand carbon has –H and $-CH_3$ substituents, of which $-CH_3$ ranks higher by sequence rule 1. The right-hand carbon has $-CH(CH_3)_2$ and $-CH_2OH$ substituents, which are equivalent by rule 1. By rule 2, however, $-CH_2OH$ ranks higher than $-CH(CH_3)_2$ because the substituent $-CH_2OH$ has an oxygen as its highest second atom, but $-CH(CH_3)_2$ has a carbon as its highest second atom. The two higher-ranked groups are on the same side of the double bond, so we assign a *Z* configuration.



***Z* configuration**

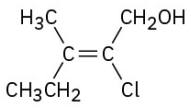
PROBLEM 7-11 Which member in each of the following sets ranks higher?

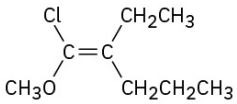
(a) –H or $-CH_3$ **(b)** –Cl or $-CH_2Cl$ **(c)** $-CH_2CH_2Br$ or $-CH{=}CH_2$ **(d)** $-NHCH_3$ or $-OCH_3$
(e) $-CH_2OH$ or $-CH{=}O$ **(f)** $-CH_2OCH_3$ or $-CH{=}O$

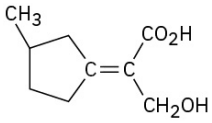
PROBLEM 7-12 Rank the substituents in each of the following sets according to the sequence rules:

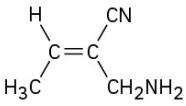
(a) $-CH_3$, –OH, –H, –Cl **(b)** $-CH_3$, $-CH_2CH_3$, $-CH{=}CH_2$, $-CH_2OH$
(c) $-CO_2H$, $-CH_2OH$, $-C{\equiv}N$, $-CH_2NH_2$ **(d)** $-CH_2CH_3$, $-C{\equiv}CH$, $-C{\equiv}N$, $-CH_2OCH_3$

PROBLEM 7-13 Assign *E* or *Z* configuration to the following alkenes:

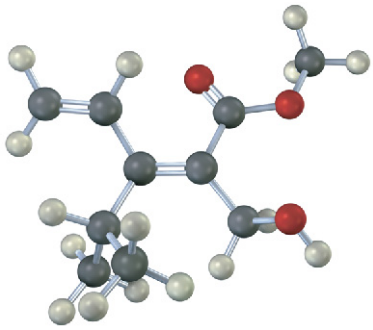
(a)


(b)


(c)


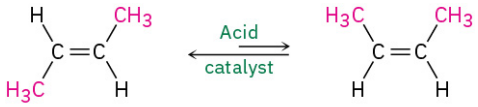
(d)


PROBLEM 7-14 Assign stereochemistry (*E* or *Z*) to the double bond in the following compound, and convert the drawing into a skeletal structure (red = O):



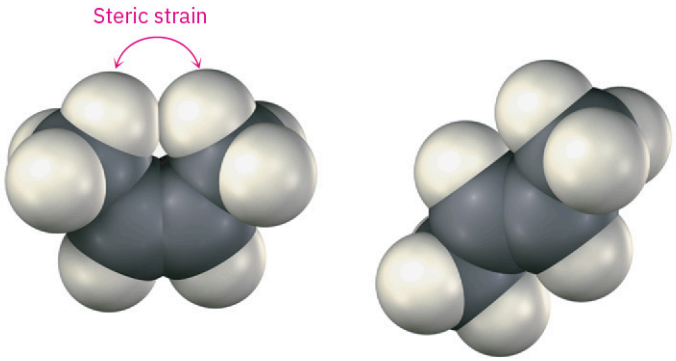
7.6 Stability of Alkenes

Although the cis–trans interconversion of alkene isomers does not occur spontaneously, it can often be made to happen by treating the alkene with a strong acid catalyst. If we interconvert *cis*-2-butene with *trans*-2-butene and allow them to reach equilibrium, we find that they aren't of equal stability. The trans isomer is more stable than the cis isomer by 2.8 kJ/mol (0.66 kcal/mol) at room temperature, corresponding to a 76 : 24 ratio.



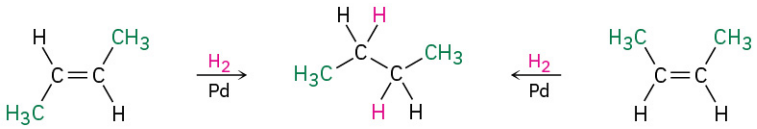
Trans (76%) **Cis (24%)**

Cis alkenes are less stable than their trans isomers because of steric strain between the two larger substituents on the same side of the double bond. This is the same kind of steric interference that we saw previously in the axial conformation of methylcyclohexane (**Section 4.7**).



***cis*-2-Butene** ***trans*-2-Butene**

Although it's sometimes possible to find relative stabilities of alkene isomers by establishing a cis–trans equilibrium through treatment with strong acid, a more general method is to take advantage of the fact that alkenes undergo a *hydrogenation* reaction to give the corresponding alkane when treated with H_2 gas in the presence of a catalyst such as palladium or platinum.



***trans*-2-Butene** **Butane** ***cis*-2-Butene**

Energy diagrams for the hydrogenation reactions of *cis*- and *trans*-2-butene are shown in **FIGURE 7.6**. Because *cis*-2-butene is less stable than *trans*-2-butene by 2.8 kJ/mol, the energy diagram shows the cis alkene at a higher energy level. After reaction, however, both curves are at the same energy level (butane). It therefore follows that $\Delta G°$ for reaction of the cis isomer must be larger than $\Delta G°$ for reaction of the trans isomer by 2.8 kJ/mol. In other words, more energy is released in the hydrogenation of the cis isomer than the trans isomer because the cis isomer has more energy to begin with.

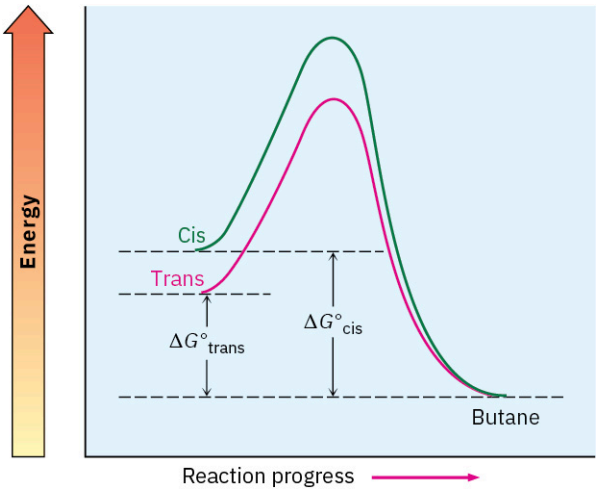
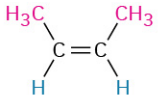
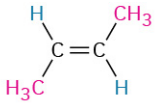


FIGURE 7.6 Energy diagrams for hydrogenation of *cis*- and *trans*-2-butene. The cis isomer is higher in energy than the trans isomer by about 2.8 kJ/mol and therefore releases more energy when hydrogenated.

If we were to measure the so-called heats of hydrogenation ($\Delta H°_{hydrog}$) for two double-bond isomers and find their difference, we could determine the relative stabilities of cis and trans isomers without having to measure an equilibrium position. *cis*-2-Butene, for instance, has $\Delta H°_{hydrog}$ = −119 kJ/mol(−28.3 kcal/mol), while *trans*-2-butene has $\Delta H°_{hydrog}$ = −115 kJ/mol(−27.4 kcal/mol)—a difference of 4 kJ/mol.



Cis isomer
$\Delta H°_{hydrog}$ = −120 kJ/mol



Trans isomer
$\Delta H°_{hydrog}$ = −116 kJ/mol

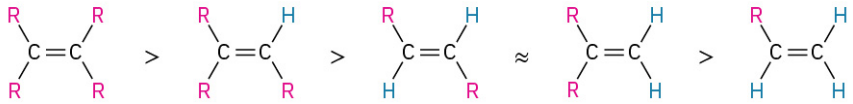
The 4 kJ/mol energy difference between the 2-butene isomers calculated from heats of hydrogenation agrees reasonably well with the 2.8 kJ/mol energy difference calculated from equilibrium data, but the values aren't exactly the same for two reasons. First, there is probably some experimental error, because heats of hydrogenation are difficult to measure accurately. Second, heats of reaction and equilibrium constants don't measure exactly the same thing. Heats of reaction measure enthalpy changes, $\Delta H°$, whereas equilibrium constants measure free-energy changes, $\Delta G°$, so we might expect a slight difference between the two.

TABLE 7.2 Heats of Hydrogenation of Some Alkenes

Substitution	Alkene	$\Delta H°_{hydrog}$ (kJ/mol)	$\Delta H°_{hydrog}$ (kcal/mol)
Ethylene	$H_2C{=}CH_2$	−136	−32.6
Monosubstituted	$CH_3CH{=}CH_2$	−125	−29.9
Disubstituted	$CH_3CH{=}CHCH_3$ (cis) $CH_3CH{=}CHCH_3$ (trans) $(CH_3)_2C{=}CH_2$	−119 −115 −118	−28.3 −27.4 −28.2
Trisubstituted	$(CH_3)_2C{=}CHCH_3$	−112	−26.7
Tetrasubstituted	$(CH_3)_2C{=}C(CH_3)_2$	−110	−26.4

TABLE 7.2 lists some representative data for the hydrogenation of different alkenes and shows that alkenes become more stable with increasing substitution. That is, alkenes follow the stability order:

Tetrasubstituted > Trisubstituted > Disubstituted > Monosubstituted



The stability order of substituted alkenes is due to a combination of two factors. One is a stabilizing interaction between the C=C π orbital and adjacent C–H σ bonds on substituents. In valence-bond language, the interaction is called **hyperconjugation**. In a molecular orbital description, there is a bonding MO that extends over the four-atom C=C–C–H grouping, as shown in FIGURE 7.7. The more substituents present on the double bond, the more hyperconjugation occurs and the more stable the alkene.

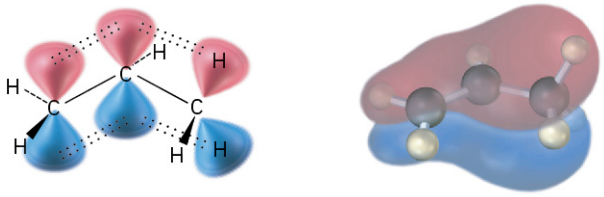


FIGURE 7.7 Hyperconjugation is a stabilizing interaction between the C=C π orbital and adjacent C–H σ bonds on substituents. The more substituents there are, the greater the stabilization of the alkene.

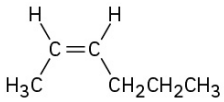
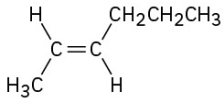
A second factor that contributes to alkene stability involves bond strengths. A bond between a sp^2 carbon and a sp^3 carbon is somewhat stronger than a bond between two sp^3 carbons. Thus, in comparing 1-butene and 2-butene, the monosubstituted isomer has one sp^3–sp^3 bond and one sp^3–sp^2 bond, while the disubstituted isomer has two sp^3–sp^2 bonds. More highly substituted alkenes always have a higher ratio of sp^3–sp^2 bonds to sp^3–sp^3 bonds than less highly substituted alkenes and are therefore more stable.

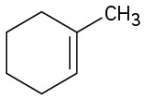
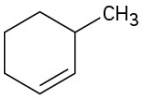


2-Butene (more stable) **1-Butene (less stable)**

PROBLEM 7-15 Name the following alkenes, and tell which compound in each pair is more stable:

(a) $H_2C{=}CHCH_2CH_3$ or $H_2C{=}C(CH_3)CH_3$

(b)  or 

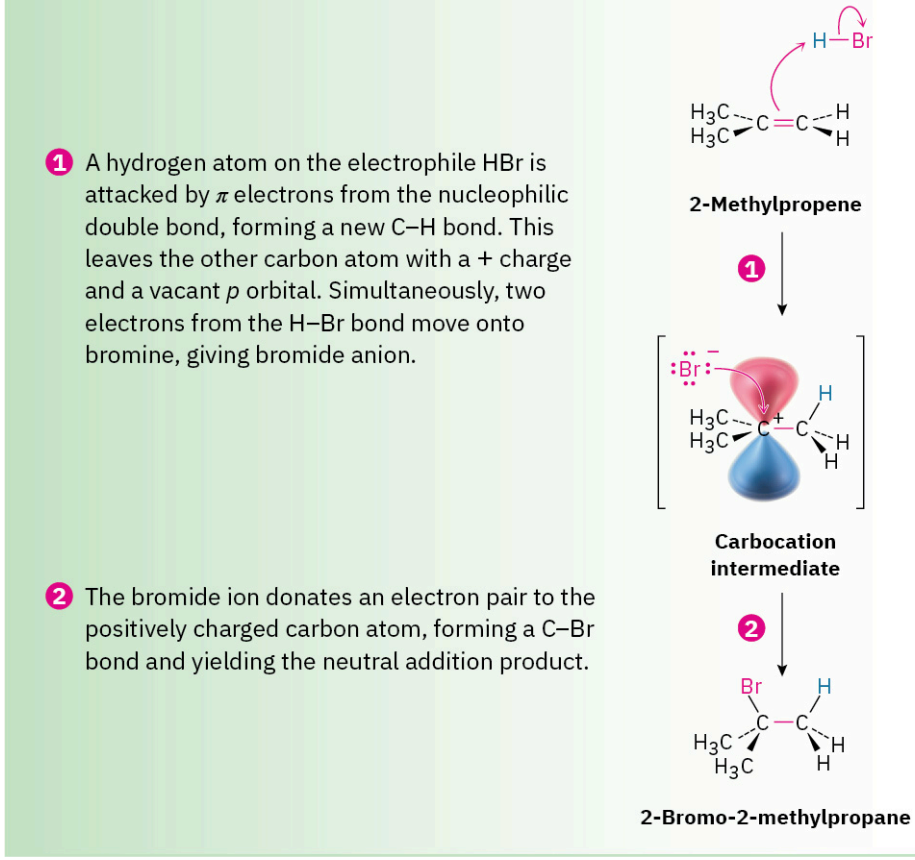
(c)  or 

7.7 Electrophilic Addition Reactions of Alkenes

Before beginning a detailed discussion of alkene reactions, let's review briefly some conclusions from the previous chapter. We said in Section 6.5 that alkenes behave as nucleophiles (Lewis bases) in polar reactions, donating a pair of electrons from their electron-rich C=C bond to an electrophile (Lewis acid). For example, reaction of 2-methylpropene with HBr yields 2-bromo-2-methylpropane. A careful study of this and similar reactions by the British chemist Christopher Ingold and others in the 1930s led to the generally accepted mechanism shown in FIGURE 7.8 for an **electrophilic addition reaction**.

FIGURE 7.8 MECHANISM

Mechanism of the electrophilic addition of HBr to 2-methylpropene. The reaction occurs in two steps, protonation and bromide addition, and involves a carbocation intermediate.



The reaction begins with an attack on the hydrogen of the electrophile HBr by the electrons of the nucleophilic π bond. Two electrons from the π bond form a new σ bond between the entering hydrogen and an alkene carbon, as shown by the curved arrow at the top of **FIGURE 7.8**. The resulting carbocation intermediate is itself an electrophile, which can accept an electron pair from nucleophilic Br^- ion to form a C–Br bond and yield a neutral addition product.

An energy diagram for the overall electrophilic addition reaction (**FIGURE 7.9**) has two peaks (transition states) separated by a valley (carbocation intermediate). The energy level of the intermediate is higher than that of the starting alkene, but the reaction as a whole is exergonic (negative $\Delta G°$). The first step, protonation of the alkene to yield the intermediate cation, is relatively slow. But once the cation intermediate is formed, it rapidly reacts to yield the final alkyl bromide product. The relative rates of the two steps are indicated in **FIGURE 7.9** by the fact that $\Delta G_1^{\ddagger}$ is larger than $\Delta G_2^{\ddagger}$.

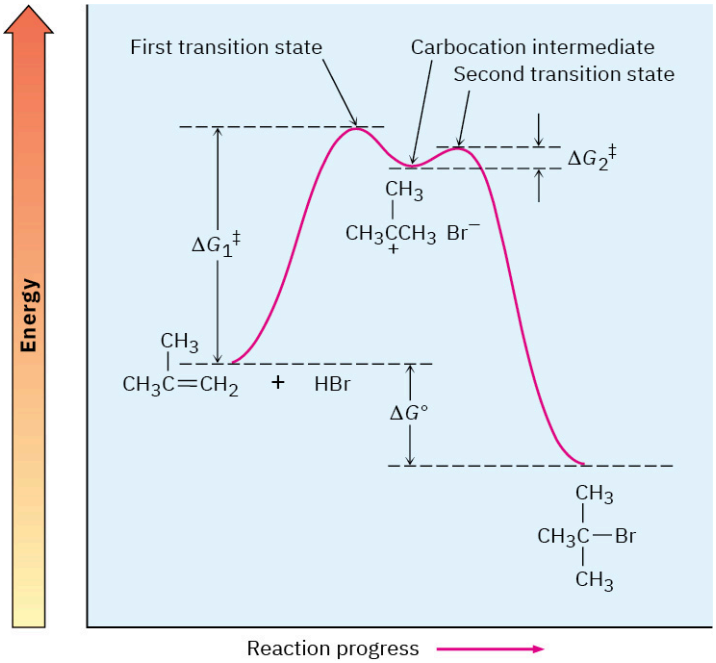
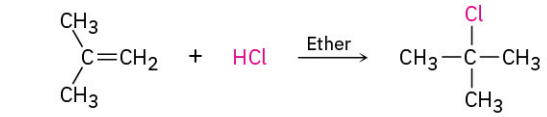
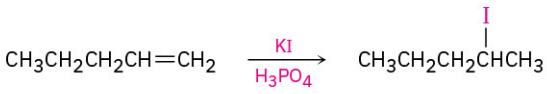


FIGURE 7.9 Energy diagram for the two-step electrophilic addition of HBr to 2-methylpropene. The first step is slower than the second step.

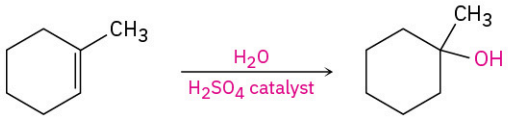
Electrophilic addition to alkenes is successful not only with HBr but with HCl, HI, and H_2O as well. Note that HI is usually generated in the reaction mixture by treating potassium iodide with phosphoric acid and that a strong acid catalyst is needed for the addition of water.



2-Methylpropene **2-Chloro-2-methylpropane (94%)**



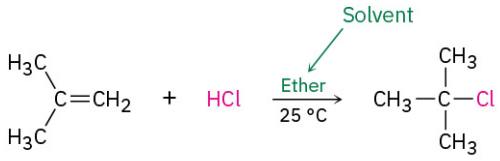
1-Pentene **(HI)** **2-Iodopentane**



1-Methylcyclohexene **1-Methylcyclohexanol**

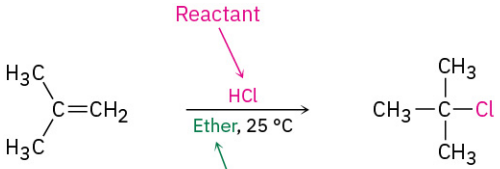
WRITING ORGANIC REACTIONS

This is a good place to mention that the equations for organic reactions are sometimes written in different ways to emphasize different points. In describing a laboratory process, for instance, the reaction of 2-methylpropene with HCl might be written in the format A + B ⟶ C to emphasize that both reactants are equally important for the purposes of the discussion. The solvent and notes about other reaction conditions such as temperature are written either above or below the reaction arrow.



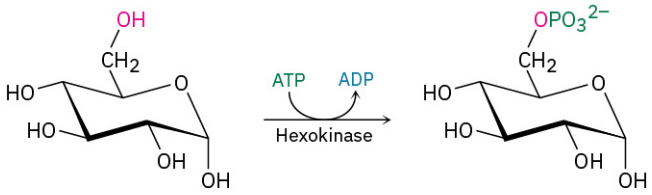
2-Methylpropene **2-Chloro-2-methyl-propane**

Alternatively, we might write the same reaction in a format to emphasize that 2-methylpropene is the reactant whose chemistry is of greater interest. The second reactant, HCl, is placed above the reaction arrow together with notes about solvent and reaction conditions.



2-Methylpropene **2-Chloro-2-methyl-propane**

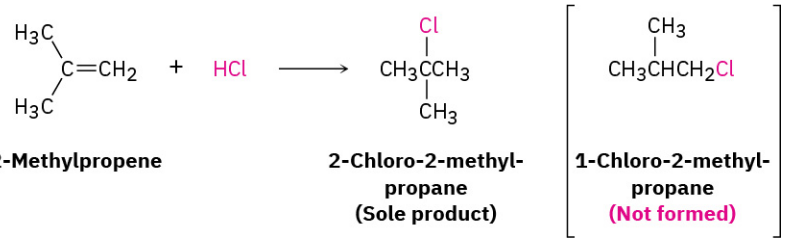
In describing a biological process, the reaction is almost always written to show only the structures of the primary reactant and product, while abbreviating the structures of various biological "reagents" and by-products with a curved arrow that intersects the straight reaction arrow. As discussed in **Section 6.11**, the reaction of glucose with ATP to give glucose 6-phosphate plus ADP would then be written as



Glucose **Glucose 6-phosphate**

7.8 Orientation of Electrophilic Additions: Markovnikov's Rule

Look carefully at the electrophilic addition reactions shown in the previous section. In each case, an unsymmetrically substituted alkene has given a single addition product rather than the mixture that might be expected. For example, 2-methylpropene might react with HCl to give both 2-chloro-2-methylpropane and 1-chloro-2-methylpropane, but it doesn't. It gives only 2-chloro-2-methylpropane as the sole product. Similarly, it's invariably the case in biological alkene addition reactions that only a single product is formed. We say that such reactions are **regiospecific** (**ree**-jee-oh-specific) when only one of two possible orientations of an addition occurs.

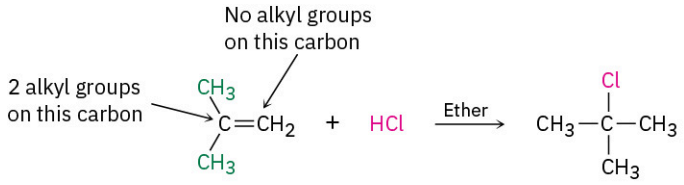


2-Methylpropene **2-Chloro-2-methyl-propane (Sole product)** **1-Chloro-2-methyl-propane (Not formed)**

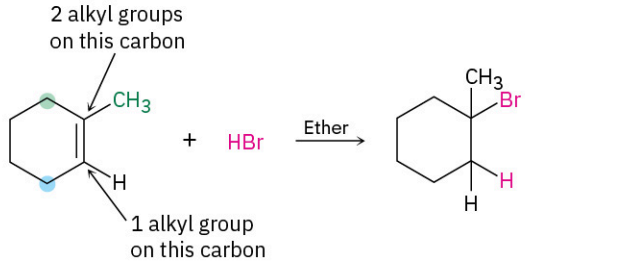
After looking at the results of many such reactions, the Russian chemist Vladimir Markovnikov proposed in 1869 what has become known as:

Markovnikov's rule

In the addition of HX to an alkene, the H attaches to the carbon with fewer alkyl substituents and the X attaches to the carbon with more alkyl substituents.

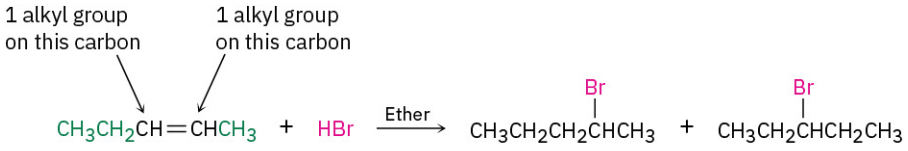


2-Methylpropene **2-Chloro-2-methylpropane**



1-Methylcyclohexene **1-Bromo-1-methylcyclohexane**

When both double-bonded carbon atoms have the same degree of substitution, a mixture of addition products results.



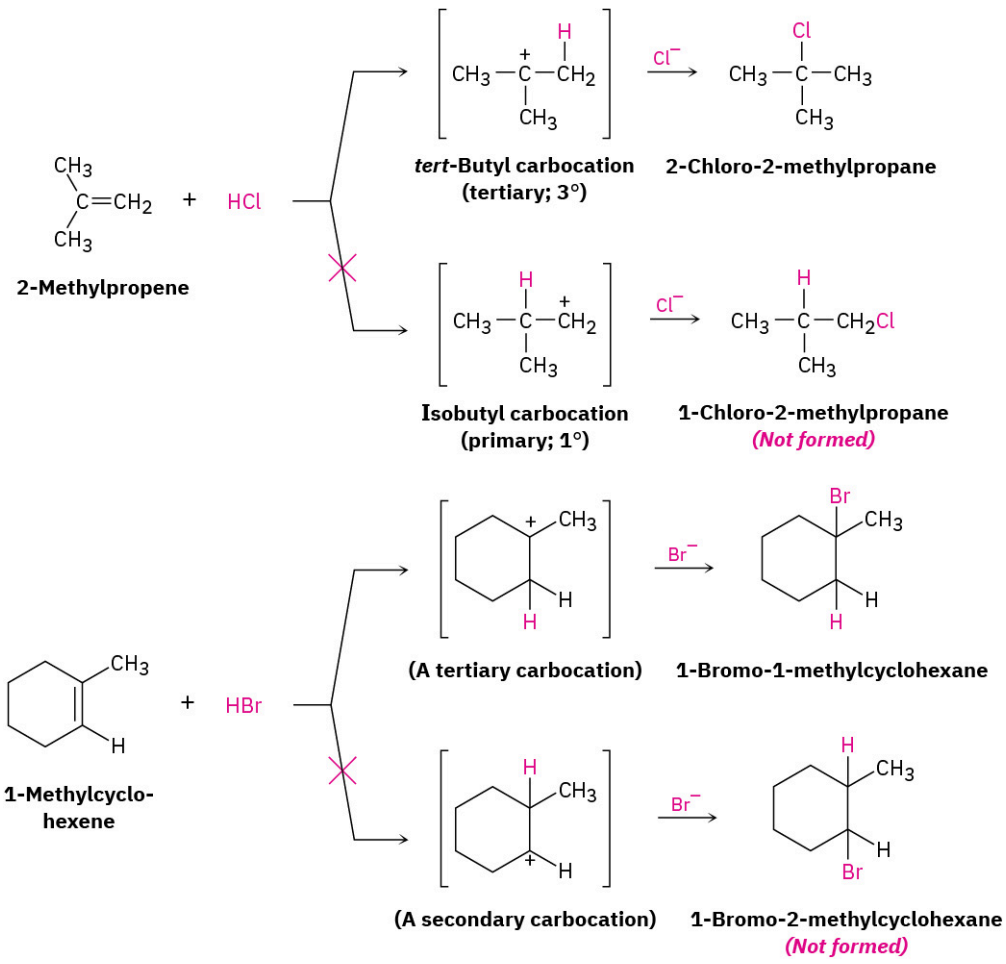
2-Pentene **2-Bromopentane** **3-Bromopentane**

Because carbocations are involved as intermediates in these electrophilic addition reactions, Markovnikov's rule can be restated in the following way:

Markovnikov's rule restated

In the addition of HX to an alkene, the more highly substituted carbocation is formed as the intermediate rather than the less highly substituted one.

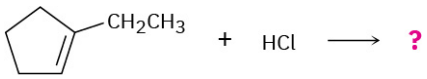
For example, addition of H^+ to 2-methylpropene yields the intermediate *tertiary* carbocation rather than the alternative primary carbocation, and addition to 1-methylcyclohexene yields a tertiary cation rather than a secondary one. Why should this be?



WORKED EXAMPLE 7.2

Predicting the Product of an Electrophilic Addition Reaction

What product would you expect from reaction of HCl with 1-ethylcyclopentene?

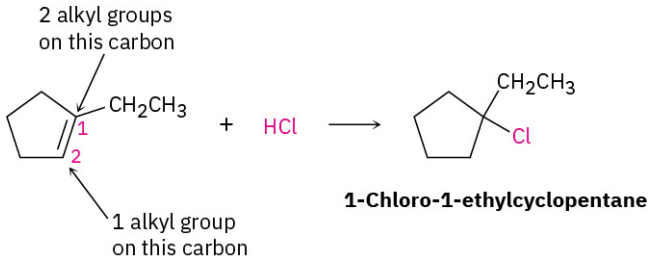


Strategy

When solving a problem that asks you to predict a reaction product, begin by looking at the functional group(s) in the reactants and deciding what kind of reaction is likely to occur. In the present instance, the reactant is an alkene that will probably undergo an electrophilic addition reaction with HCl. Next, recall what you know about electrophilic addition reactions to predict the product. You know that electrophilic addition reactions follow Markovnikov's rule, so H^+ will add to the double-bond carbon that has one alkyl group (C2 on the ring) and the Cl will add to the double-bond carbon that has two alkyl groups (C1 on the ring).

Solution

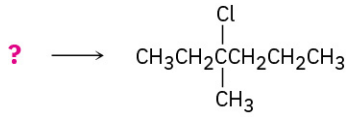
The expected product is 1-chloro-1-ethylcyclopentane.



WORKED EXAMPLE 7.3

Synthesizing a Specific Compound

What alkene would you start with to prepare the following alkyl halide? There may be more than one possibility.

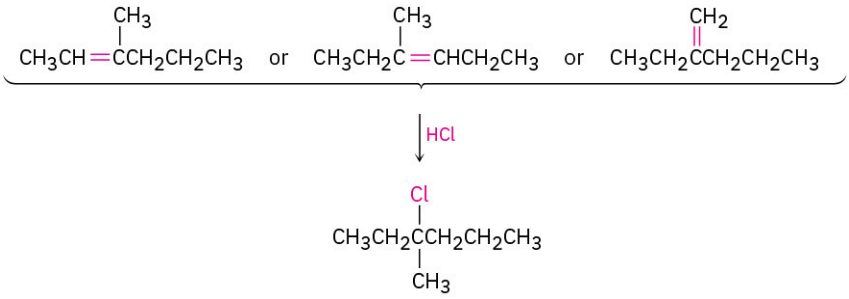


Strategy

When solving a problem that asks how to prepare a given product, always work backward. Look at the product, identify the functional group(s) it contains, and ask yourself, "How can I prepare that functional group?" In the present instance, the product is a tertiary alkyl chloride, which can be prepared by reaction of an alkene with HCl. The carbon atom bearing the –Cl atom in the product must be one of the double-bond carbons in the reactant. Draw and evaluate all possibilities.

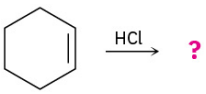
Solution

There are three possibilities, all of which could give the desired product according to Markovnikov's rule.

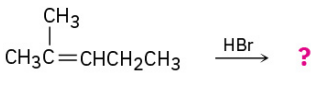


PROBLEM 7-16 Predict the products of the following reactions:

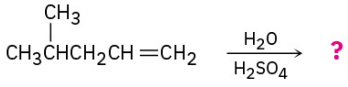
(a)



(b)

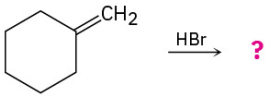


(c)



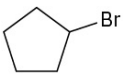
(Addition of H_2O occurs.)

(d)

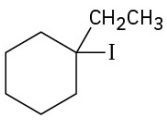


PROBLEM 7-17 What alkenes would you start with to prepare the following products?

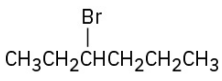
(a)



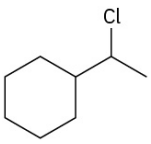
(b)



(c)



(d)



7.9 Carbocation Structure and Stability

To understand why Markovnikov's rule works, we need to learn more about the structure and stability of carbocations and about the general nature of reactions and transition states. The first point to explore involves structure.

A great deal of experimental evidence has shown that carbocations are planar. The trivalent carbon is sp^2-hybridized, and the three substituents are oriented toward the corners of an equilateral triangle, as indicated in **FIGURE 7.10**. Because there are only six valence electrons on carbon and all six are used in the three σ bonds, the p orbital extending above and below the plane is unoccupied.

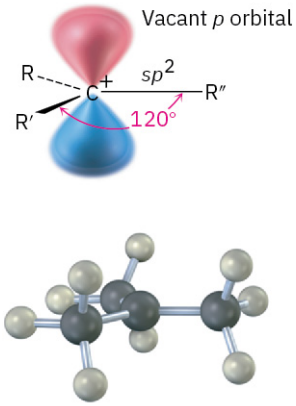
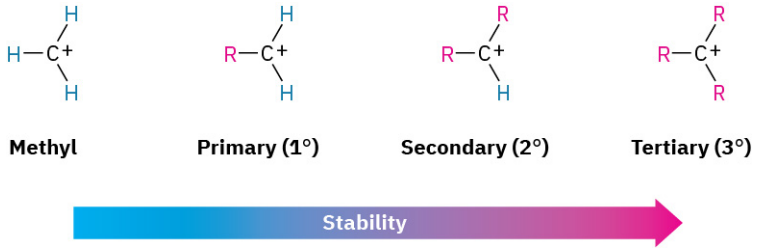


FIGURE 7.10 The structure of a carbocation. The trivalent carbon is sp^2-hybridized and has a vacant p orbital perpendicular to the plane of the carbon and three attached groups.

The second point to explore involves carbocation stability. 2-Methylpropene might react with H^+ to form a carbocation having three alkyl substituents (a tertiary ion, 3°), or it might react to form a carbocation having one alkyl substituent (a primary ion, 1°). Since the tertiary alkyl chloride, 2-chloro-2-methylpropane, is the only product observed, formation of the tertiary cation is evidently favored over formation of the primary cation. Thermodynamic measurements show that, indeed, the stability of carbocations increases with increasing substitution so that the stability order is tertiary > secondary > primary > methyl.



One way of determining carbocation stabilities is to measure the amount of energy required to form a carbocation by dissociation of the corresponding alkyl halide, $R - X \rightarrow R^+ + :X^-$. As shown in **FIGURE 7.11**, tertiary alkyl halides dissociate to give carbocations more easily than secondary or primary ones. Thus, trisubstituted carbocations are more stable than disubstituted ones, which are more stable than monosubstituted ones. The data in **FIGURE 7.11** are taken from measurements made in the gas phase, but a similar stability order is found for carbocations in solution. The dissociation enthalpies are much lower in solution because polar solvents can stabilize the ions, but the order of carbocation stability remains the same.

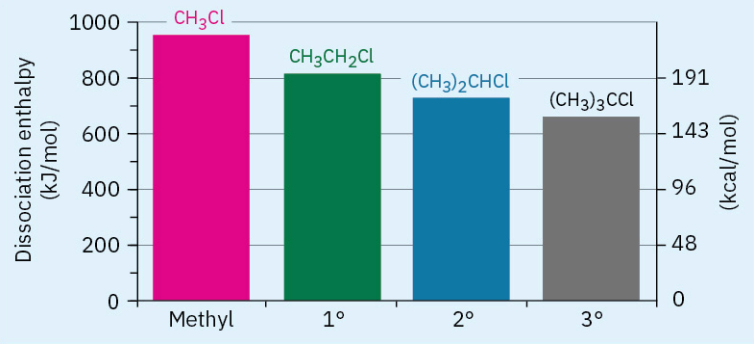


FIGURE 7.11 A plot of dissociation enthalpy versus substitution pattern for the gas-phase dissociation of alkyl chlorides to yield carbocations. More highly substituted alkyl halides dissociate more easily than less highly substituted ones.

Why are more highly substituted carbocations more stable than less highly substituted ones? There are at least two reasons. Part of the answer has to do with inductive effects, and part has to do with hyperconjugation. Inductive effects, discussed in **Section 2.1** in connection with polar covalent bonds, result from the shifting of electrons in a σ bond in response to the electronegativity of nearby atoms. In the present instance, electrons from a relatively larger and more polarizable alkyl group can shift toward a neighboring positive charge more easily than the electron from a hydrogen. Thus, the more alkyl groups attached to the positively charged carbon, the more electron density shifts toward the charge and the more inductive stabilization of the cation occurs (**FIGURE 7.12**).

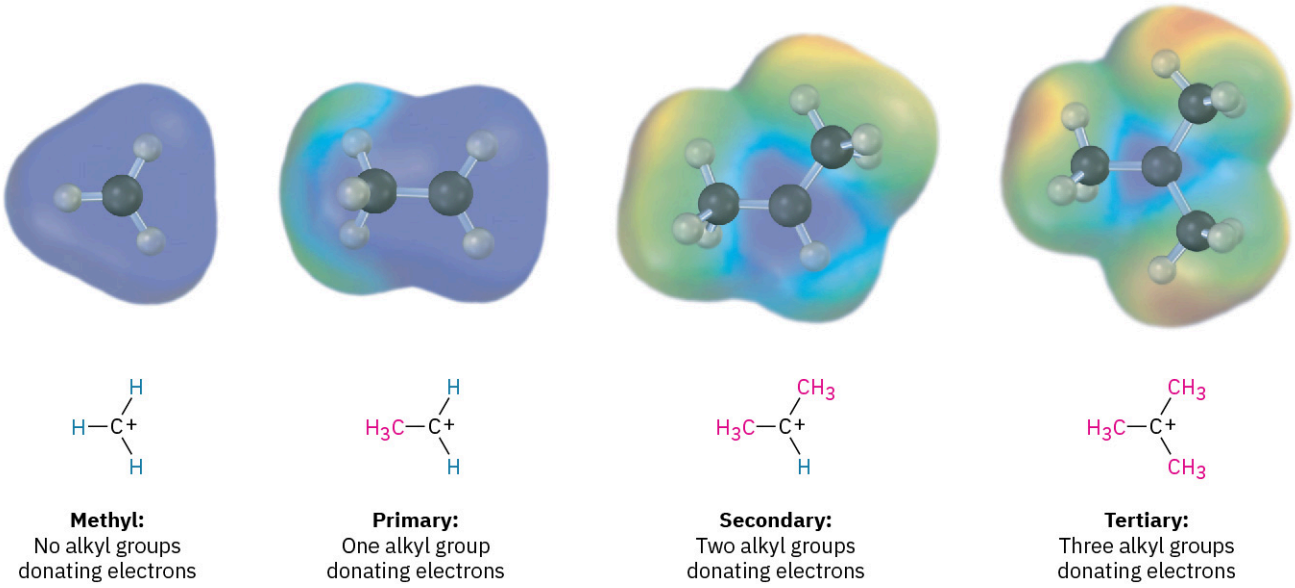


FIGURE 7.12 A comparison of inductive stabilization for methyl, primary, secondary, and tertiary carbocations. The more alkyl groups that are bonded to the positively charged carbon, the more electron density shifts toward the charge, making the charged carbon less electron-poor (blue in electrostatic potential maps).

Hyperconjugation, discussed in **Section 7.6** in connection with the stabilities of substituted alkenes, is the stabilizing interaction between a p orbital and properly oriented C–H σ bonds on neighboring carbons that are roughly parallel to the p orbital. The more alkyl groups there are on the carbocation, the more possibilities there are for hyperconjugation and the more stable the carbocation. **FIGURE 7.13** shows the molecular orbital for the ethyl carbocation, $CH_3CH_2^+$, and indicates the difference between the C–H bond perpendicular to the cation p orbital and the two C–H bonds more parallel to the cation p orbital. Only these roughly parallel C–H bonds are oriented properly to take part in hyperconjugation.

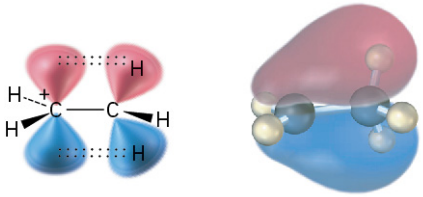
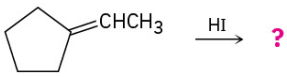


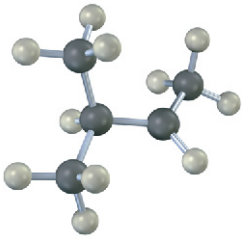
FIGURE 7.13 **Stabilization of the ethyl carbocation, $CH_3CH_2^+$, through hyperconjugation**. Interaction of neighboring C–H σ bonds with the vacant *p* orbital stabilizes the cation and lowers its energy. The molecular orbital shows that only the two C–H bonds more parallel to the cation *p* orbital are oriented properly. The C–H bond perpendicular to the cation *p* orbital can't take part properly.

PROBLEM 7-18 Show the structures of the carbocation intermediates you would expect in the following reactions:

(a) $CH_3CH_2C(CH_3)=CHCH_3 \xrightarrow{HBr}$?

(b) 

PROBLEM 7-19 Draw a skeletal structure of the following carbocation. Identify it as primary, secondary, or tertiary, and identify the hydrogen atoms that have the proper orientation for hyperconjugation in the conformation shown.



7.10 The Hammond Postulate

Let's summarize what we've learned of electrophilic addition reactions to this point:

- **Electrophilic addition to an unsymmetrically substituted alkene gives the more substituted carbocation intermediate.** A more substituted carbocation forms faster than a less substituted one and, once formed, rapidly goes on to give the final product.
- **A more substituted carbocation is more stable than a less substituted one.** That is, the stability order of carbocations is tertiary > secondary > primary > methyl.

What we have not yet seen is how these two points are related. Why does the stability of the carbocation intermediate affect the rate at which it's formed and thereby determine the structure of the final product? After all, carbocation stability is determined by the free-energy change $\Delta G°$, but reaction rate is determined by the activation energy $\Delta G^{\ddagger}$. The two quantities aren't directly related.

Although there is no simple quantitative relationship between the stability of a carbocation intermediate and the rate of its formation, there *is* an intuitive relationship. It's generally true when comparing two similar reactions that the more stable intermediate forms faster than the less stable one. The situation is shown graphically in FIGURE 7.14, where the energy profile in part **(a)** represents the typical situation, as opposed to the profile in part **(b)**. That is, the curves for two similar reactions don't cross one another.

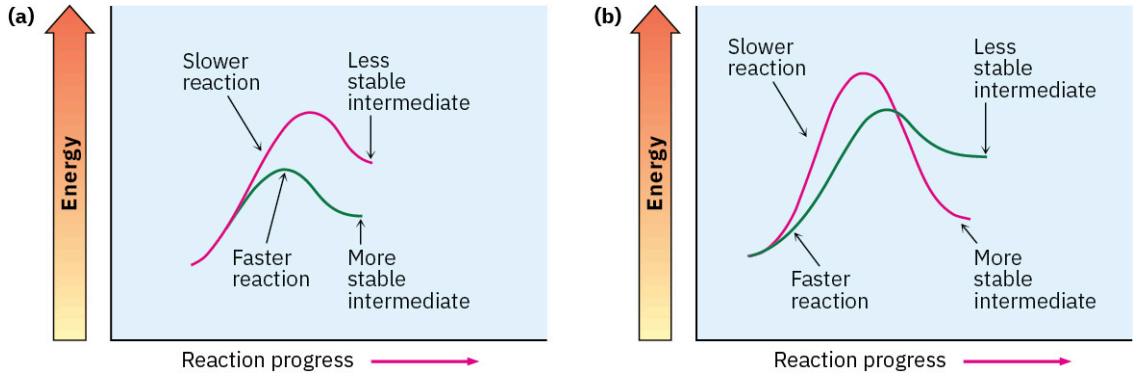


FIGURE 7.14 Energy diagrams for two similar competing reactions. In **(a)**, the faster reaction yields the more stable intermediate. In **(b)**, the slower reaction yields the more stable intermediate. The curves shown in **(a)** represent the typical situation.

Called the **Hammond postulate**, the explanation of the relationship between reaction rate and intermediate stability goes like this: Transition states represent energy maxima. They are high-energy activated complexes that occur transiently during the course of a reaction and immediately go on to a more stable species. Although we can't actually observe transition states because they have no finite lifetime, the Hammond postulate says that we can get an idea of a particular transition state's structure by looking at the structure of the nearest stable species. Imagine the two cases shown in **FIGURE 7.15**, for example. The reaction profile in part **(a)** shows the energy curve for an endergonic reaction step, and the profile in part **(b)** shows the curve for an exergonic step.

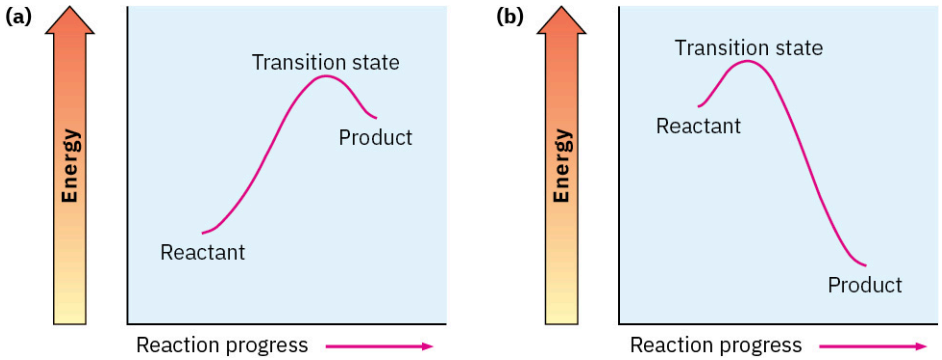


FIGURE 7.15 Energy diagrams for endergonic and exergonic steps. **(a)** In an endergonic step, the energy levels of transition state and *product* are closer. **(b)** In an exergonic step, the energy levels of transition state and *reactant* are closer.

In an endergonic reaction (**FIGURE 7.15**a), the energy level of the transition state is closer to that of the product than that of the reactant. Since the transition state is closer energetically to the product, we make the natural assumption that it's also closer structurally. In other words, the transition state for an endergonic reaction step structurally resembles the product of that step. Conversely, the transition state for an exergonic reaction (**FIGURE 7.15**b), is closer energetically, and thus structurally, to the reactant than to the product. We therefore say that the transition state for an exergonic reaction step structurally resembles the reactant for that step.

Hammond postulate:
The structure of a transition state resembles the structure of the nearest stable species. Transition states for endergonic steps structurally resemble products, and transition states for exergonic steps structurally resemble reactants.

How does the Hammond postulate apply to electrophilic addition reactions? The formation of a carbocation by protonation of an alkene is an endergonic step. Thus, the transition state for alkene protonation structurally resembles the carbocation intermediate, and any factor that stabilizes the carbocation will also stabilize the nearby transition state. Since increasing alkyl substitution stabilizes carbocations, it also stabilizes the transition states leading to those ions, thus resulting in a faster reaction. In other words, more stable carbocations form faster because their greater stability is reflected in the lower-energy transition state leading to them (**FIGURE 7.16**).

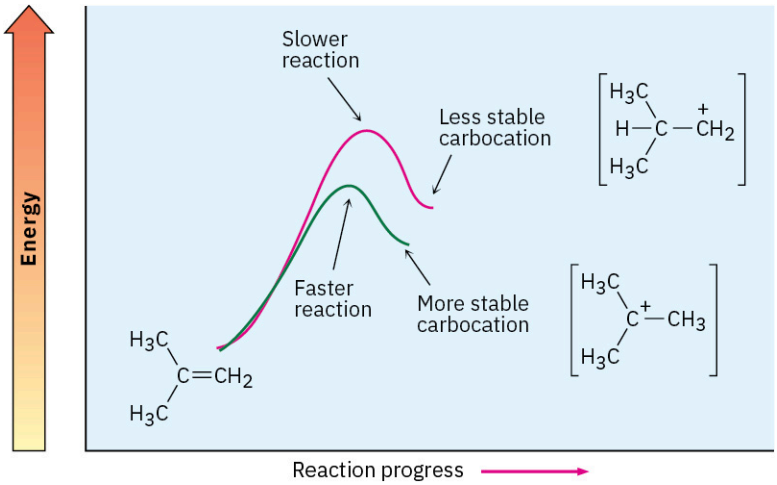


FIGURE 7.16 Energy diagrams for carbocation formation. The more stable tertiary carbocation is formed faster **(green curve)** because its increased stability lowers the energy of the transition state leading to it.

We can imagine the transition state for alkene protonation to be a structure in which one of the alkene carbon atoms has almost completely rehybridized from sp^2 to sp^3 and the remaining alkene carbon bears much of the positive charge (**FIGURE 7.17**). This transition state is stabilized by hyperconjugation and inductive effects in the same way as the product carbocation. The more alkyl groups that are present, the greater the extent of stabilization and the faster the transition state forms.

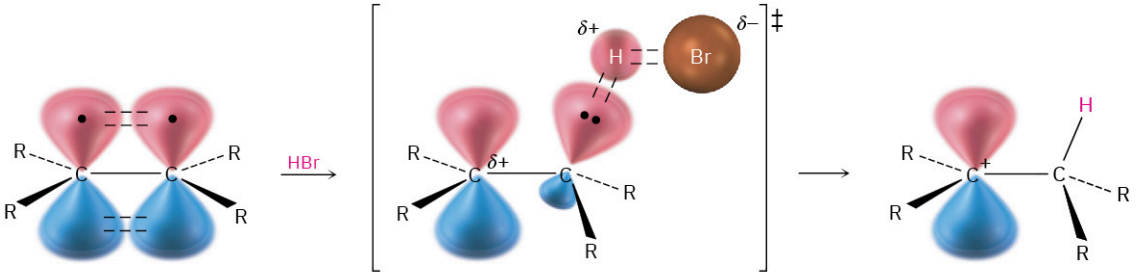


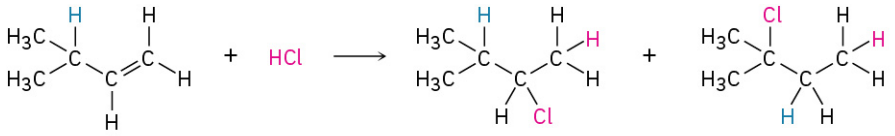
FIGURE 7.17 The hypothetical structure of a transition state for alkene protonation. The transition state is closer in both energy and structure to the carbocation than to the alkene. Thus, an increase in carbocation stability (lower $\Delta G°$) also causes an increase in transition-state stability (lower $\Delta G^{‡}$), thereby increasing the rate of its formation.

PROBLEM 7-20 What about the second step in the electrophilic addition of HCl to an alkene—the reaction of chloride ion with the carbocation intermediate? Is this step exergonic or endergonic? Does the transition state for this second step resemble the reactant (carbocation) or product (alkyl chloride)? Make a rough drawing of what the transition-state structure might look like.

7.11 Evidence for the Mechanism of Electrophilic Additions: Carbocation Rearrangements

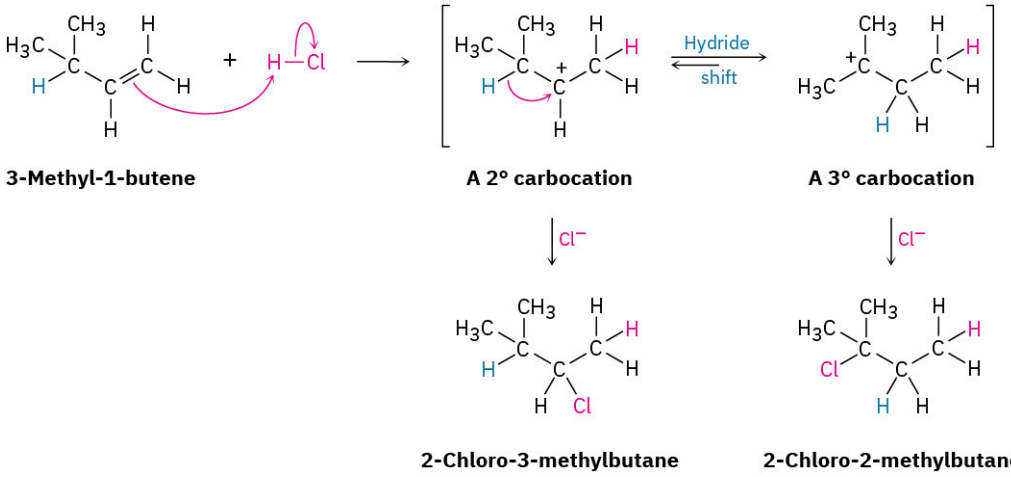
How do we know that the carbocation mechanism for electrophilic addition reactions of alkenes is correct? The answer is that we *don't* know it's correct; at least we don't know with complete certainty. Although an incorrect reaction mechanism can be disproved by demonstrating that it doesn't account for observed data, a correct reaction mechanism can never be entirely proven. The best we can do is to show that a proposed mechanism is consistent with all known facts. If enough facts are accounted for, the mechanism is probably correct.

One of the best pieces of evidence supporting the carbocation mechanism for the electrophilic addition reaction was discovered during the 1930s by F. C. Whitmore of Pennsylvania State University, who found that structural rearrangements often occur during the reaction of HX with an alkene. For example, reaction of HCl with 3-methyl-1-butene yields a substantial amount of 2-chloro-2-methylbutane in addition to the "expected" product, 2-chloro-3-methylbutane.



3-Methyl-1-butene **2-Chloro-3-methylbutane (approx. 50%)** **2-Chloro-2-methylbutane (approx. 50%)**

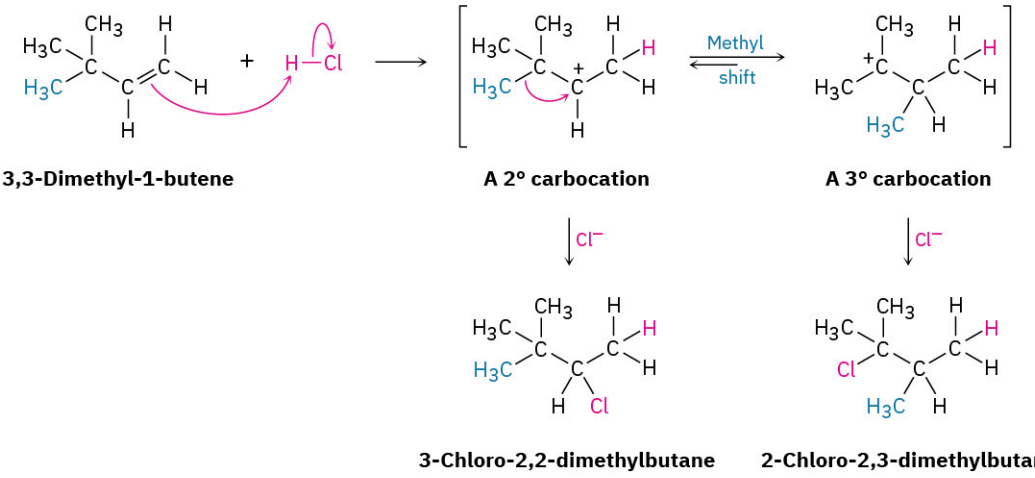
If the reaction takes place in a single step, it would be difficult to account for rearrangement, but if the reaction takes place in several steps, rearrangement is more easily explained. Whitmore suggested that it is a carbocation intermediate that undergoes rearrangement. The secondary carbocation intermediate formed by protonation of 3-methyl-1-butene rearranges to a more stable tertiary carbocation by a **hydride shift**—the shift of a hydrogen atom and its electron pair (a hydride ion, **:**H^-) between neighboring carbons.



3-Methyl-1-butene **A 2° carbocation** **A 3° carbocation**

2-Chloro-3-methylbutane **2-Chloro-2-methylbutane**

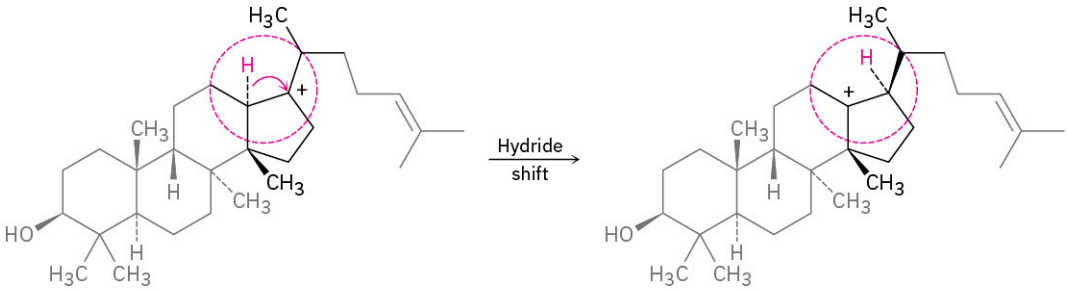
Carbocation rearrangements can also occur by the shift of an alkyl group with its electron pair. For example, reaction of 3,3-dimethyl-1-butene with HCl leads to an equal mixture of unrearranged 3-chloro-2,2-dimethylbutane and rearranged 2-chloro-2,3-dimethylbutane. In this instance, a secondary carbocation rearranges to a more stable tertiary carbocation by the shift of a methyl group.



3,3-Dimethyl-1-butene **A 2° carbocation** **A 3° carbocation**

3-Chloro-2,2-dimethylbutane **2-Chloro-2,3-dimethylbutane**

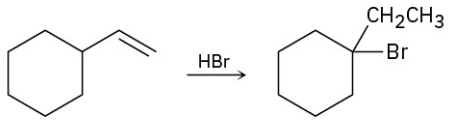
Note the similarities between the two carbocation rearrangements: in both cases, a group (:H^- or :CH_3^-) moves to an adjacent positively charged carbon, taking its bonding electron pair with it. Also in both cases, a less stable carbocation rearranges to a more stable ion. Rearrangements of this kind are a common feature of carbocation chemistry and are particularly important in the biological pathways by which steroids and related substances are synthesized. An example is the following hydride shift that occurs during the biosynthesis of cholesterol.



A tertiary carbocation **An isomeric tertiary carbocation**

As always, when looking at any complex chemical transformation, whether biochemical or not, focus on the part of the molecule where the change is occurring and don't worry about the rest. The tertiary carbocation just pictured looks complicated, but all the chemistry is taking place in the small part of the molecule inside the red circle.

PROBLEM 7-21 On treatment with HBr, vinylcyclohexane undergoes addition and rearrangement to yield 1-bromo-1-ethylcyclohexane. Using curved arrows, propose a mechanism to account for this result.



Vinylcyclohexane **1-Bromo-1-ethylcyclohexane**

CHEMISTRY MATTERS

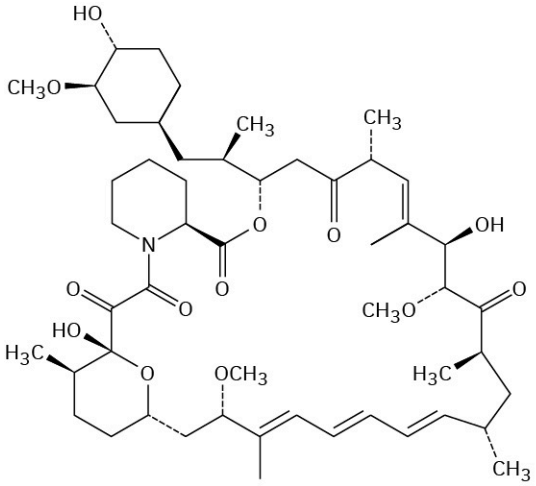
Bioprospecting: Hunting for Natural Products

Most people know the names of the common classes of biomolecules—proteins, carbohydrates, lipids, and nucleic acids—but there are many more kinds of compounds in living organisms than just those four. All living organisms also contain a vast diversity of substances usually grouped under the heading *natural products*. The term **natural product** really refers to any naturally occurring substance but is generally taken to mean a so-called secondary metabolite—a small molecule that is not essential to the growth and development of the producing organism and is not classified by structure.



FIGURE 7.18 Rapamycin, an immunosuppressant natural product used during organ transplants, was originally isolated from a soil sample found on Rapa Nui (Easter Island), an island 2200 miles off the coast of Chile known for its giant Moai statues. (credit: modification of work "Moai facing inland at Ahu Tongariki" by Ian Sewell/Wikimedia Commons, CC BY 2.5)

It has been estimated that well over 300,000 secondary metabolites exist, and it's thought that their primary function is to increase the likelihood of an organism's survival by repelling or attracting other organisms. Alkaloids, such as morphine; antibiotics, such as erythromycin and the penicillins; and immunosuppressive agents, such as rapamycin (sirolimus) prescribed for liver transplant recipients, are examples.



Rapamycin (Sirolimus)

Where do these natural products come from, and how are they found? Although most chemists and biologists spend their working time in the laboratory, a few spend their days scuba diving on South Pacific islands or trekking through the rainforests of South America and Southeast Asia at work as bioprospectors. Their job is to

hunt for new and unusual natural products that might be useful as drugs.

As noted in the **Chapter 6 *Chemistry Matters***, more than half of all new drug candidates come either directly or indirectly from natural products. Morphine from the opium poppy, prostaglandin E_1 from sheep prostate glands, erythromycin A from a *Streptomyces erythreus* bacterium cultured from a Philippine soil sample, and benzylpenicillin from the mold *Penicillium notatum* are examples. The immunosuppressive agent rapamycin, whose structure is shown previously, was first isolated from a *Streptomyces hygroscopicus* bacterium found in a soil sample from Rapa Nui (Easter Island), located 2200 miles off the coast of Chile.

With less than 1% of living organisms yet investigated, bioprospectors have a lot of work to do. But there is a race going on. Rainforests throughout the world are being destroyed at an alarming rate, causing many species of both plants and animals to become extinct before they can even be examined. The governments in many countries seem aware of the problem, but there is as yet no international treaty on biodiversity that could help preserve vanishing species.

Key Terms

- **alkene ($R_2C{=}CR_2$)**
- **allyl group**
- **degree of unsaturation**
- ***E* configuration**
- ***E,Z* system**
- **electrophilic addition reaction**
- **Hammond postulate**
- **hydride shift**
- **hyperconjugation**
- **Markovnikov's rule**
- **methylene group**
- **regiospecific reaction**
- **unsaturated**
- **vinyl group**
- ***Z* configuration**

Summary

Carbon–carbon double bonds are present in most organic and biological molecules, so a good understanding of their behavior is needed. In this chapter, we've looked at some consequences of alkene stereoisomerism and at the details of the broadest class of alkene reactions—the electrophilic addition reaction.

An **alkene** is a hydrocarbon that contains a carbon–carbon double bond. Because they contain fewer hydrogens than alkanes with the same number of carbons, alkenes are said to be **unsaturated**.

Because rotation around the double bond can't occur, substituted alkenes can exist as cis–trans stereoisomers. The configuration of a double bond can be specified by applying the Cahn–Ingold–Prelog sequence rules, which rank the substituents on each double-bond carbon. If the higher-ranking groups on each carbon are on the same side of the double bond, the configuration is ***Z*** (*zusammen,* "together"); if the higher-ranking groups on each carbon are on opposite sides of the double bond, the configuration is ***E*** (*entgegen,* "apart").

Alkene chemistry is dominated by **electrophilic addition reactions**. When HX reacts with an unsymmetrically substituted alkene, **Markovnikov's rule** predicts that the H will add to the carbon having fewer alkyl substituents and the X group will add to the carbon having more alkyl substituents. Electrophilic additions to alkenes take place through carbocation intermediates formed by reaction of the nucleophilic alkene π bond with electrophilic H^+. Carbocation stability follows the order

Tertiary (3°) > Secondary (2°) > Primary (1°) > Methyl

$R_3C^+ > R_2CH^+ > RCH_2{}^+ > CH_3{}^+$

Markovnikov's rule can be restated by saying that, in the addition of HX to an alkene, a more stable carbocation intermediate is formed. This result is explained by the **Hammond postulate**, which says that the transition state of an exergonic reaction step structurally resembles the reactant, whereas the transition state of an endergonic reaction step structurally resembles the product. Since an alkene protonation step is endergonic, the stability of the more highly substituted carbocation is reflected in the stability of the transition state leading to its formation.

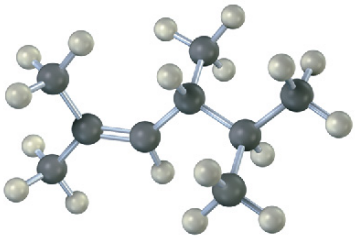
Evidence in support of a carbocation mechanism for electrophilic additions comes from the observation that structural rearrangements often take place during reaction. Rearrangements occur by shift of either a hydride ion, **:**H^- (a **hydride shift**), or an alkyl anion, **:**R^-, from a carbon atom to the neighboring positively charged carbon. This results in isomerization of a less stable carbocation to a more stable one.

Additional Problems

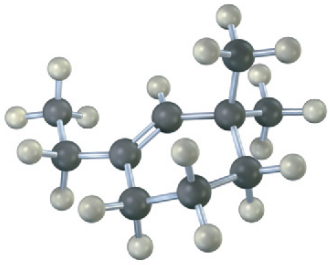
Visualizing Chemistry

PROBLEM 7-22 Name the following alkenes, and convert each drawing into a skeletal structure:

(a)

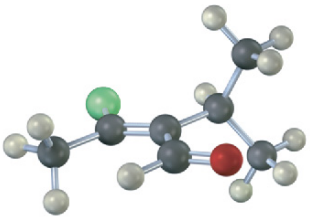


(b)

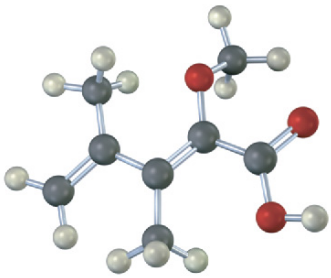


PROBLEM 7-23 Assign *E* or *Z* stereochemistry to the double bonds in each of the following alkenes, and convert each drawing into a skeletal structure (red = O, green = Cl):

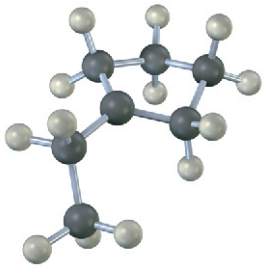
(a)



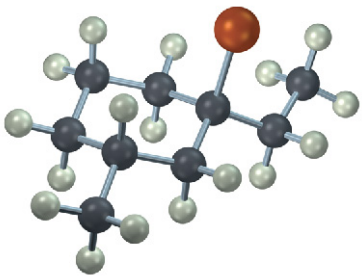
(b)



PROBLEM 7-24 The following carbocation is an intermediate in the electrophilic addition reaction of HCl with two different alkenes. Identify both, and tell which C–H bonds in the carbocation are aligned for hyperconjugation with the vacant *p* orbital on the positively charged carbon.

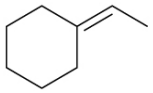


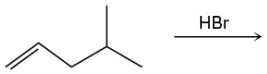
PROBLEM 7-25 The following alkyl bromide can be made by HBr addition to three different alkenes. Show their structures.

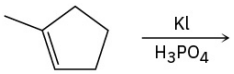


Mechanism Problems

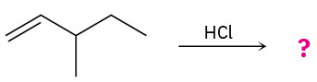
PROBLEM 7-26 Predict the major product and show the complete mechanism for each of the following electrophilic addition reactions.

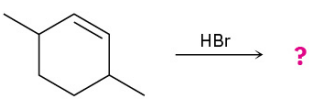
(a) 

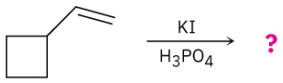
(b) 

(c) 

PROBLEM 7-27 Each of the following electrophilic addition reactions involves a carbocation rearrangement. Predict the product and draw the complete mechanism of each using curved arrows.

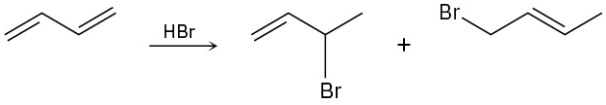
(a) 

(b) 

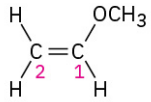
(c) 

two different rearrangement products

PROBLEM 7-28 When 1,3-butadiene reacts with 1 mol of HBr, two isolable products result. Propose mechanisms for both.

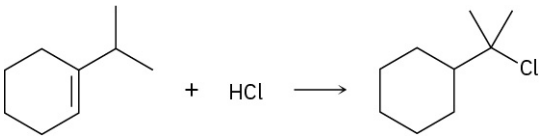


PROBLEM 7-29 When methyl vinyl ether reacts with a strong acid, H^+ adds to C2 instead of C1 or the oxygen atom. Explain.

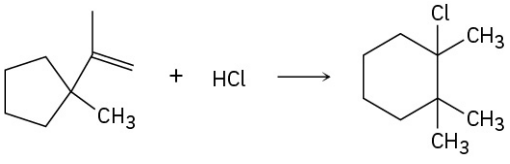


Methyl vinyl ether

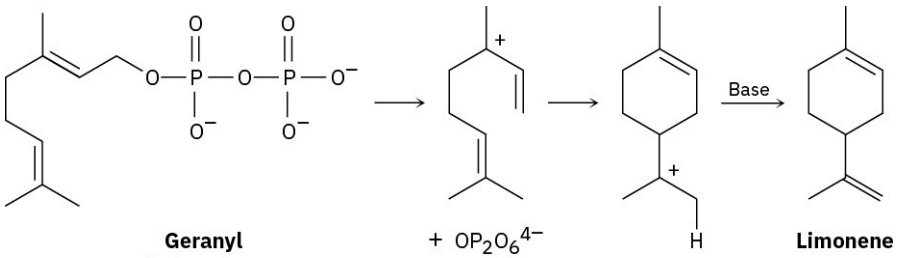
PROBLEM 7-30 Addition of HCl to 1-isopropylcyclohexene yields a rearranged product. Propose a mechanism, showing the structures of the intermediates and using curved arrows to indicate electron flow in each step.



PROBLEM 7-31 Addition of HCl to 1-isopropenyl-1-methylcyclopentane yields 1-chloro-1,2,2-trimethylcyclohexane. Propose a mechanism, showing the structures of the intermediates and using curved arrows to indicate electron flow in each step.

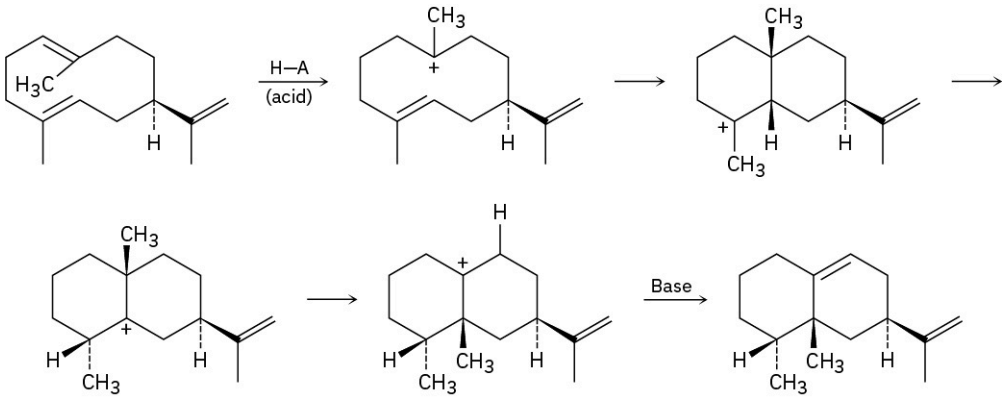


PROBLEM 7-32 Limonene, a fragrant hydrocarbon found in lemons and oranges, is biosynthesized from geranyl diphosphate by the following pathway. Add curved arrows to show the mechanism of each step. Which step involves an alkene electrophilic addition? (The ion $OP_2O_6^{4-}$ is the diphosphate ion, and "Base" is an unspecified base in the enzyme that catalyzes the reaction.)



Geranyl diphosphate **Limonene**

PROBLEM 7-33 *epi*-Aristolochene, a hydrocarbon found in both pepper and tobacco, is biosynthesized by the following pathway. Add curved arrows to show the mechanism of each step. Which steps involve alkene electrophilic addition(s), and which involve carbocation rearrangement(s)? (The abbreviation H—A stands for an unspecified acid, and "Base" is an unspecified base in the enzyme.)



***epi*-Aristolochene**

Calculating a Degree of Unsaturation

PROBLEM 7-34 Calculate the degree of unsaturation in the following formulas, and draw five possible structures for each:

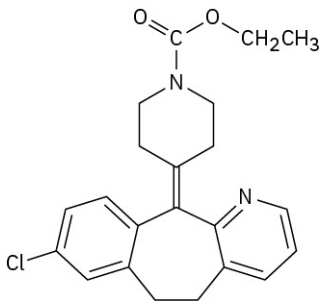
(a) $C_{10}H_{16}$ **(b)** C_8H_8O **(c)** $C_7H_{10}Cl_2$ **(d)** $C_{10}H_{16}O_2$ **(e)** $C_5H_9NO_2$ **(f)** $C_8H_{10}ClNO$

PROBLEM 7-35 How many hydrogens does each of the following compounds have?

(a) $C_8H_?O_2$, has two rings and one double bond **(b)** $C_7H_?N$, has two double bonds

(c) $C_9H_?NO$, has one ring and three double bonds

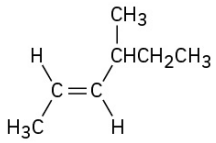
PROBLEM 7-36 Loratadine, marketed as an antiallergy medication under the brand name Claritin, has four rings, eight double bonds, and the formula $C_{22}H_?ClN_2O_2$. How many hydrogens does loratadine have? (Calculate your answer; don't count hydrogens in the structure.)

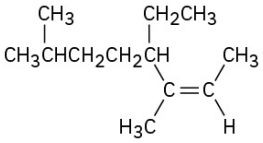


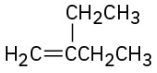
Loratadine

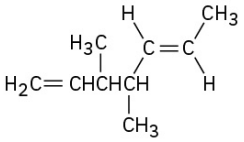
Naming Alkenes

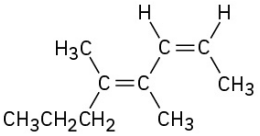
PROBLEM 7-37 Name the following alkenes:

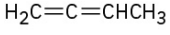
(a) 

(b) 

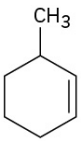
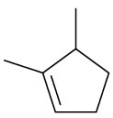
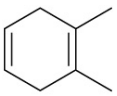
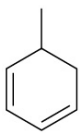
(c) 

(d) 

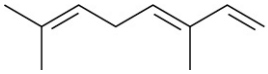
(e) 

(f) 

PROBLEM 7-38 Draw structures corresponding to the following systematic names:
(a) (4*E*)-2,4-Dimethyl-1,4-hexadiene **(b)** *cis*-3,3-Dimethyl-4-propyl-1,5-octadiene
(c) 4-Methyl-1,2-pentadiene **(d)** (3*E*,5*Z*)-2,6-Dimethyl-1,3,5,7-octatetraene
(e) 3-Butyl-2-heptene **(f)** *trans*-2,2,5,5-Tetramethyl-3-hexene

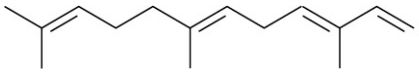
PROBLEM 7-39 Name the following cycloalkenes:
(a)  **(b)**  **(c)**  **(d)**  **(e)**  **(f)** 

PROBLEM 7-40 Ocimene is a triene found in the essential oils of many plants. What is its IUPAC name, including stereochemistry?



Ocimene

PROBLEM 7-41 α-Farnesene is a constituent of the natural wax found on apples. What is its IUPAC name, including stereochemistry?



α-Farnesene

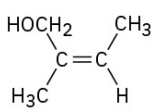
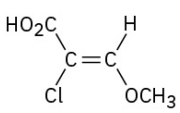
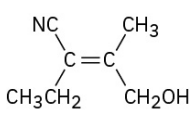
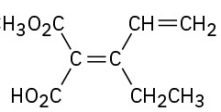
PROBLEM 7-42 Menthene, a hydrocarbon found in mint plants, has the systematic name 1-isopropyl-4-methylcyclohexene. Draw its structure.

PROBLEM 7-43 Draw and name the six alkene isomers, C_5H_{10}, including *E,Z* isomers.

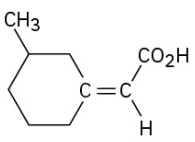
PROBLEM 7-44 Draw and name the 17 alkene isomers, C_6H_{12}, including *E,Z* isomers.

Alkene Isomers and Their Stability

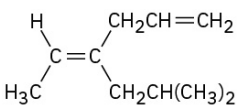
PROBLEM 7-45 Rank the following sets of substituents according to the Cahn–Ingold–Prelog sequence rules:
(a) $-CH_3$, $-Br$, $-H$, $-I$ **(b)** $-OH$, $-OCH_3$, $-H$, $-CO_2H$ **(c)** $-CO_2H$, $-CO_2CH_3$, $-CH_2OH$, $-CH_3$
(d) $-CH_3$, $-CH_2CH_3$, $-CH_2CH_2OH$, $-CCH_3$ (O) **(e)** $-CH{=}CH_2$, $-CN$, $-CH_2NH_2$, $-CH_2Br$
(f) $-CH{=}CH_2$, $-CH_2CH_3$, $-CH_2OCH_3$, $-CH_2OH$

PROBLEM 7-46 Assign *E* or *Z* configuration to each of the following compounds:
(a)  **(b)**  **(c)**  **(d)** 

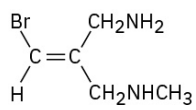
PROBLEM 7-47 Which of the following *E,Z* designations are correct, and which are incorrect?

(a) 

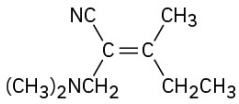
Z

(b) 

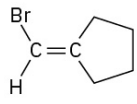
E

(c) 

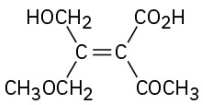
Z

(d) 

E

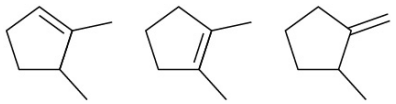
(e) 

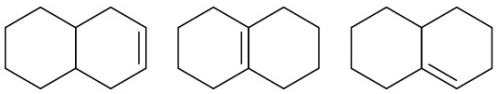
Z

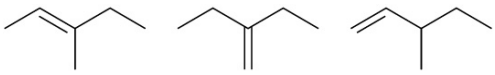
(f) 

E

PROBLEM 7-48 Rank the double bonds according to their increasing stability.

(a) 

(b) 

(c) 

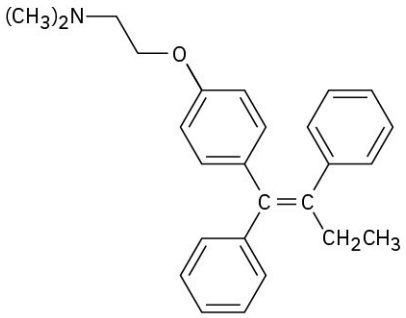
PROBLEM 7-49 *trans*-2-Butene is more stable than *cis*-2-butene by only 4 kJ/mol, but *trans*-2,2,5,5-tetramethyl-3-hexene is more stable than its cis isomer by 39 kJ/mol. Explain.

PROBLEM 7-50 Cyclodecene can exist in both cis and trans forms, but cyclohexene cannot. Explain.

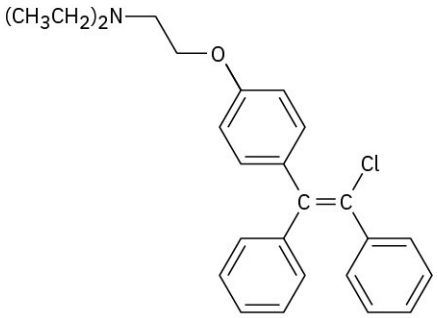
PROBLEM 7-51 Normally, a trans alkene is more stable than its cis isomer. *trans*-Cyclooctene, however, is less stable than *cis*-cyclooctene by 38.5 kJ/mol. Explain.

PROBLEM 7-52 *trans*-Cyclooctene is less stable than *cis*-cyclooctene by 38.5 kJ/mol, but *trans*-cyclononene is less stable than *cis*-cyclononene by only 12.2 kJ/mol. Explain.

PROBLEM 7-53 Tamoxifen, a drug used in the treatment of breast cancer, and clomiphene, a drug used in fertility treatment, have similar structures but very different effects. Assign *E* or *Z* configuration to the double bonds in both compounds.



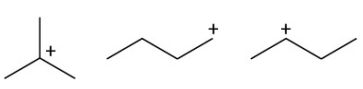
Tamoxifen (anticancer)

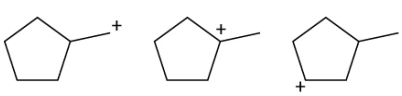


Clomiphene (fertility treatment)

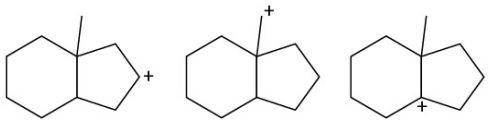
Carbocations and Electrophilic Addition Reactions

PROBLEM 7-54 Rank the following carbocations according to their increasing stability.

(a) 

(b) 

(c)



PROBLEM 7-55 Use the Hammond Postulate to determine which alkene in each pair would be expected to form a carbocation faster in an electrophilic addition reaction.

(a)



(b)

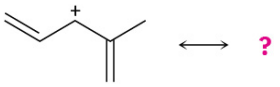


(c)

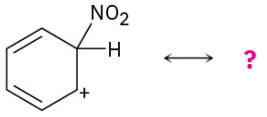


PROBLEM 7-56 The following carbocations can be stabilized by resonance. Draw all the resonance forms that would stabilize each carbocation.

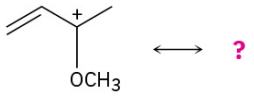
(a)



(b)

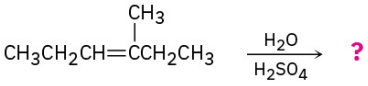


(c)



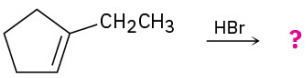
PROBLEM 7-57 Predict the major product in each of the following reactions:

(a)

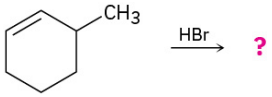


(Addition of H_2O occurs.)

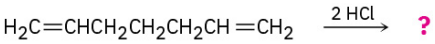
(b)



(c)

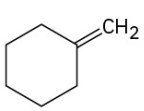


(d)

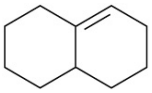


PROBLEM 7-58 Predict the major product from addition of HBr to each of the following alkenes:

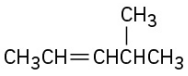
(a)



(b)

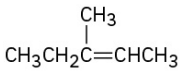


(c)

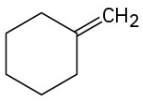


PROBLEM 7-59 Alkenes can be converted into alcohols by acid-catalyzed addition of water. Assuming that Markovnikov's rule is valid, predict the major alcohol product from each of the following alkenes.

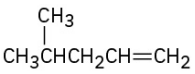
(a)



(b)

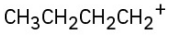


(c)

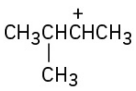


PROBLEM 7-60 Each of the following carbocations can rearrange to a more stable ion. Propose structures for the likely rearrangement products.

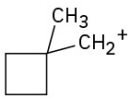
(a)



(b)



(c)

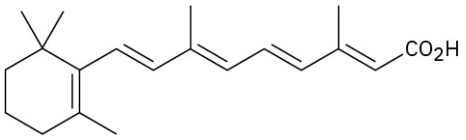


General Problems

PROBLEM 7-61 Allene (1,2-propadiene), $H_2C{=}C{=}CH_2$, has two adjacent double bonds. What kind of hybridization must the central carbon have? Sketch the bonding π orbitals in allene. What shape do you predict for allene?

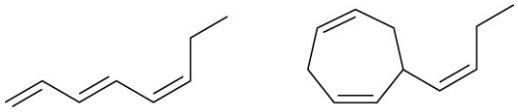
PROBLEM 7-62 The heat of hydrogenation for allene (Problem 7-61) to yield propane is −295 kJ/mol, and the heat of hydrogenation for a typical monosubstituted alkene, such as propene, is −125 kJ/mol. Is allene more stable or less stable than you might expect for a diene? Explain.

PROBLEM 7-63 Retin A, or retinoic acid, is a medication commonly used to reduce wrinkles and treat severe acne. How many different isomers arising from *E,Z* double-bond isomerizations are possible?



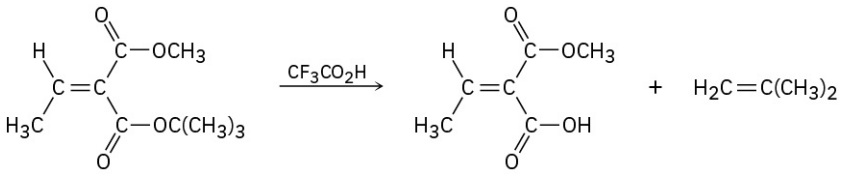
Retin A (retinoic acid)

PROBLEM 7-64 Fucoserratene and ectocarpene are sex pheromones produced by marine brown algae. What are their systematic names? (Ectocarpene is difficult; make your best guess, and then check your answer in the *Student Solutions Manual*.)

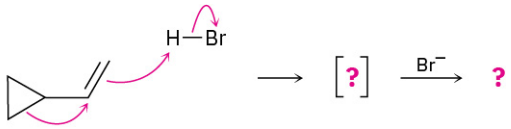


Fucoserratene **Ectocarpene**

PROBLEM 7-65 *tert*-Butyl esters [$RCO_2C(CH_3)_3$] are converted into carboxylic acids (RCO_2H) by reaction with trifluoroacetic acid, a reaction useful in protein synthesis **(Section 26.7)**. Assign *E,Z* designation to the double bonds of both reactant and product in the following scheme, and explain why there is an apparent change in double-bond stereochemistry:



PROBLEM 7-66 Vinylcyclopropane reacts with HBr to yield a rearranged alkyl bromide. Follow the flow of electrons as represented by the curved arrows, show the structure of the carbocation intermediate in brackets, and show the structure of the final product.



Vinylcyclopropane

PROBLEM 7-67 Calculate the degree of unsaturation in each of the following formulas:
(a) Cholesterol, $C_{27}H_{46}O$ **(b)** DDT, $C_{14}H_9Cl_5$ **(c)** Prostaglandin E_1, $C_{20}H_{34}O_5$
(d) Caffeine, $C_8H_{10}N_4O_2$ **(e)** Cortisone, $C_{21}H_{28}O_5$ **(f)** Atropine, $C_{17}H_{23}NO_3$

PROBLEM 7-68 The isobutyl cation spontaneously rearranges to the *tert*-butyl cation by a hydride shift. Is the rearrangement exergonic or endergonic? Draw what you think the transition state for the hydride shift might look like according to the Hammond postulate.



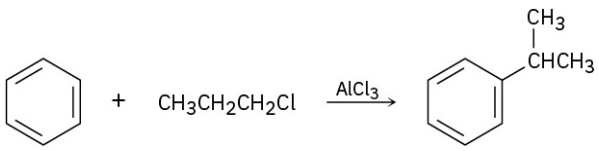
Isobutyl cation ***tert*-Butyl cation**

PROBLEM 7-69 Draw an energy diagram for the addition of HBr to 1-pentene. Let one curve on your diagram show the formation of 1-bromopentane product and another curve on the same diagram show the formation of 2-bromopentane product. Label the positions for all reactants, intermediates,

and products. Which curve has the higher-energy carbocation intermediate? Which curve has the higher-energy first transition state?

PROBLEM 7-70 Sketch the transition-state structures involved in the reaction of HBr with 1-pentene (Problem 7-69). Tell whether each structure resembles reactant or product.

PROBLEM 7-71 Aromatic compounds such as benzene react with alkyl chlorides in the presence of $AlCl_3$ catalyst to yield alkylbenzenes. This reaction occurs through a carbocation intermediate, formed by reaction of the alkyl chloride with $AlCl_3$ ($R{-}Cl + AlCl_3 \longrightarrow R^+ + AlCl_4^-$). How can you explain the observation that reaction of benzene with 1-chloropropane yields isopropylbenzene as the major product?



PROBLEM 7-72 Reaction of 2,3-dimethyl-1-butene with HBr leads to an alkyl bromide, $C_6H_{13}Br$. On treatment of this alkyl bromide with KOH in methanol, elimination of HBr occurs and a hydrocarbon that is isomeric with the starting alkene is formed. What is the structure of this hydrocarbon, and how do you think it is formed from the alkyl bromide?